CHAPTER ONE

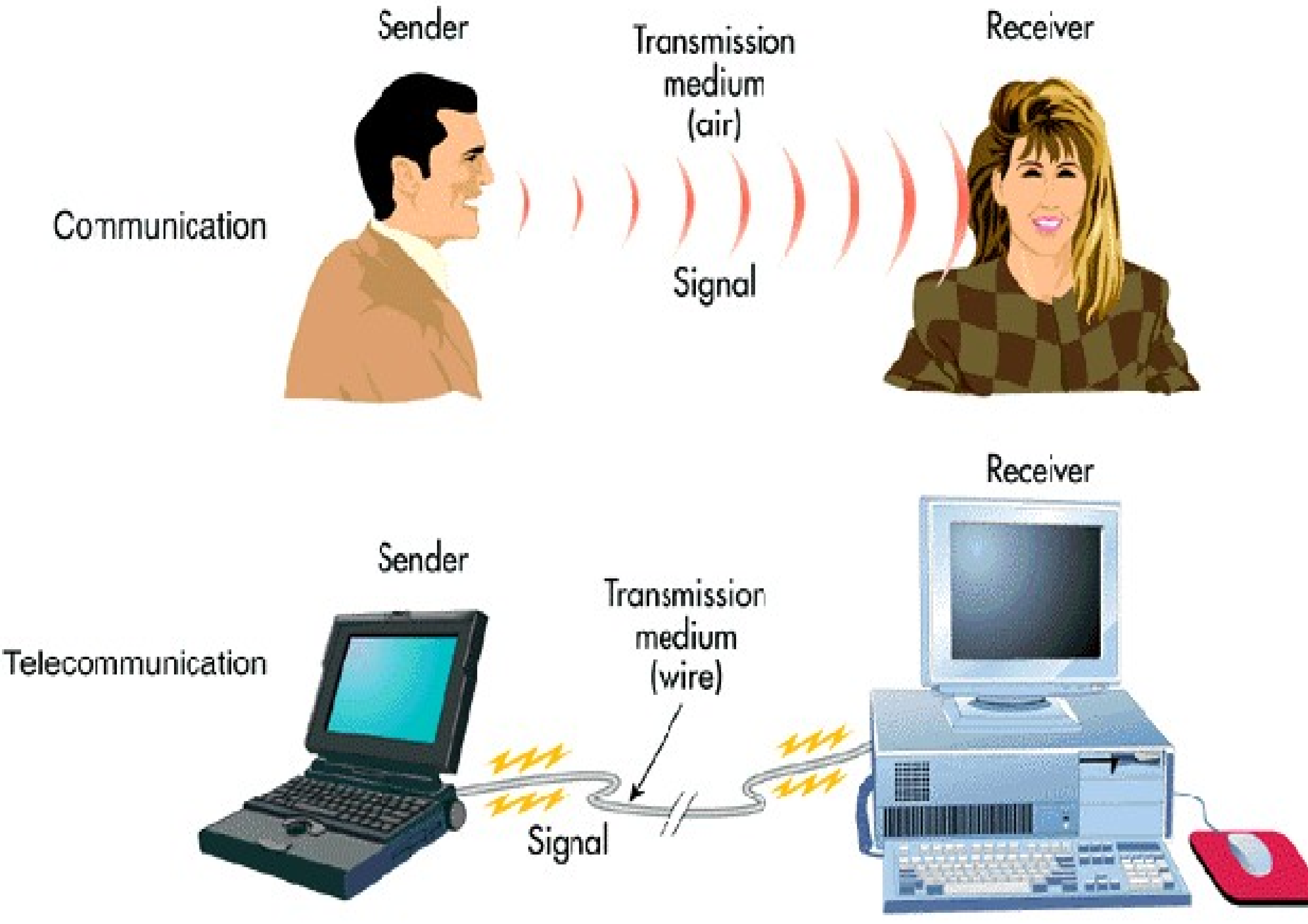
Introduction to Data communication

What is Data Communication?

- The word data refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- Data communication:- is the exchange of data between two or more devices using some form of wired or wireless transmission medium.
- Data communication is the process of transferring messages from one point to another.

Cont.…

- Electronic communications, like emails and instant messages, as well as phone calls are examples of data communications.
- A communication system can be defined as the collection of hardware and software that facilitates intersystem exchange of information between different devices.
- When we communicate, we are sharing information.
- This sharing can be local (face-to-face communication) or it may be remote or (communication over distance or telecommunication).

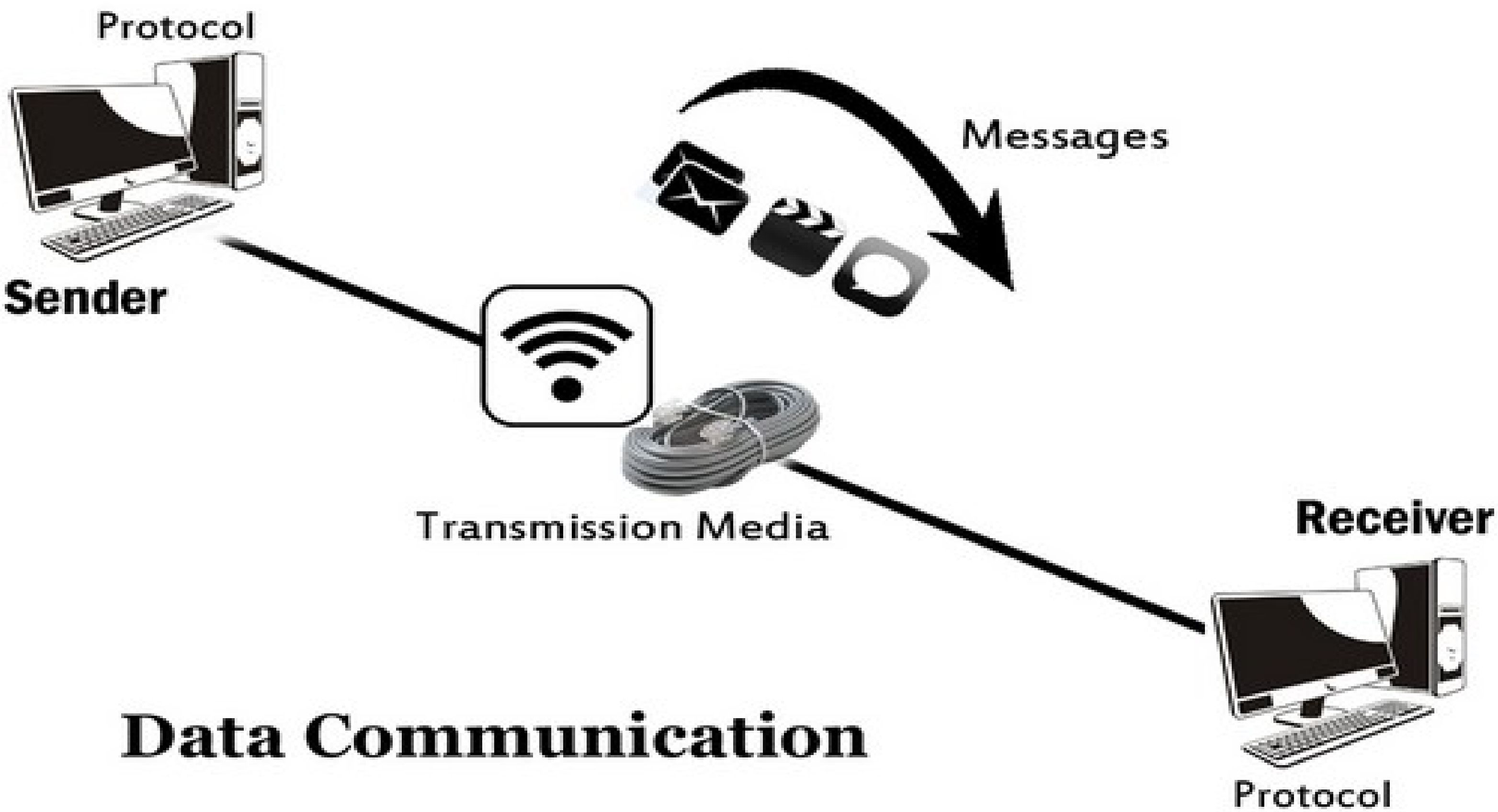

Sender
Transmission
medium
(air)
Receiver
Communication
Signal
Receiver
Sender
Transmission
medium
(wire)
Telecommunication
Signal

Components of data communication

- A data communication system is made up of five components.
  - ✓**Message:-** The message is the information (data) to be communicated.
    - ➢ It can consist of text, numbers, pictures, sound, or video- or any combination of these.
  - ✓**Sender:-** The **Sender** is the device that sends the data message.
  - ✓**Receiver:-** The **Receiver** is the device that receives the data message.
  - ✓**Medium:-**The **transmission Medium is** the physical path by which a message travels form sender to receiver.
  - ✓**A Protocol:-** is a set of rules that govern data communication.
    - ➢It represents an agreement between the communication devices. Without a protocol, two devices may be connected but not communicating.

Characteristics of Data Communication

- **Delivery:-**The system must deliver data to the correct destination.
- **Accuracy:-** The system must deliver data accurately.
- **Timeliness:-**The system must deliver data in a timely manner. Data delivered late are useless.
- **Jitter:-** Jitter refers to the variation in the packet arrival time.


Protocol
Sender
Messages
Transmission Media
Receiver
Protocol
Data Communication

Cont…

- ***Semantics*** refers to the meaning of each section of bits.
- How is a particular pattern to be interpreted, and what action is to be taken based on that interpretation?
- For example, does an address identify the route to be taken or the final destination of the message?
- ***Timing*** refers to two characteristics: when data should be sent and how fast they can be sent.
- For example, if a sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps, the transmission will overload the receiver and some data will be lost.

Data Representation

- Data representation is the method by which information is structured and stored in digital systems.
- In the age of data-driven decision-making, understanding how data is represented is critical to ensuring
  - Accuracy
  - Efficiency
  - usability.

Data Representation

- Data representation lays the foundation for all forms of computation, as it enables computers to process and manipulate data in a structured way.

Data transmission

Data transmission

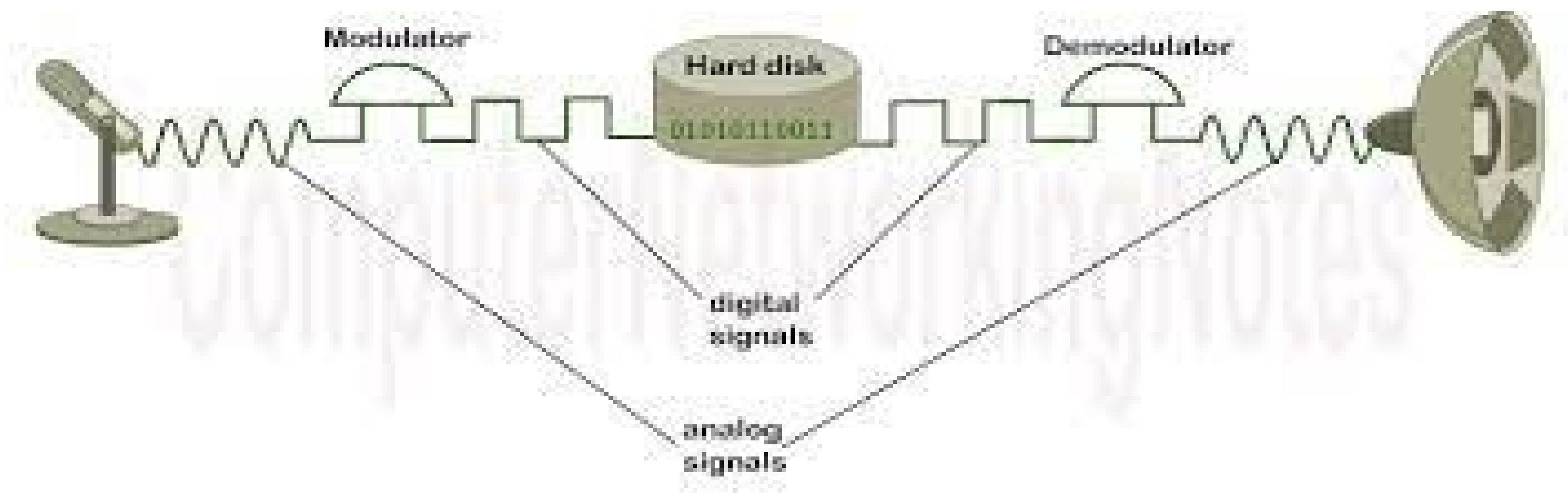
Analog data/signal and digital data/signal

- ***Analog Signal:*** - The transfer of data in the form of electrical signals or continuous waves.

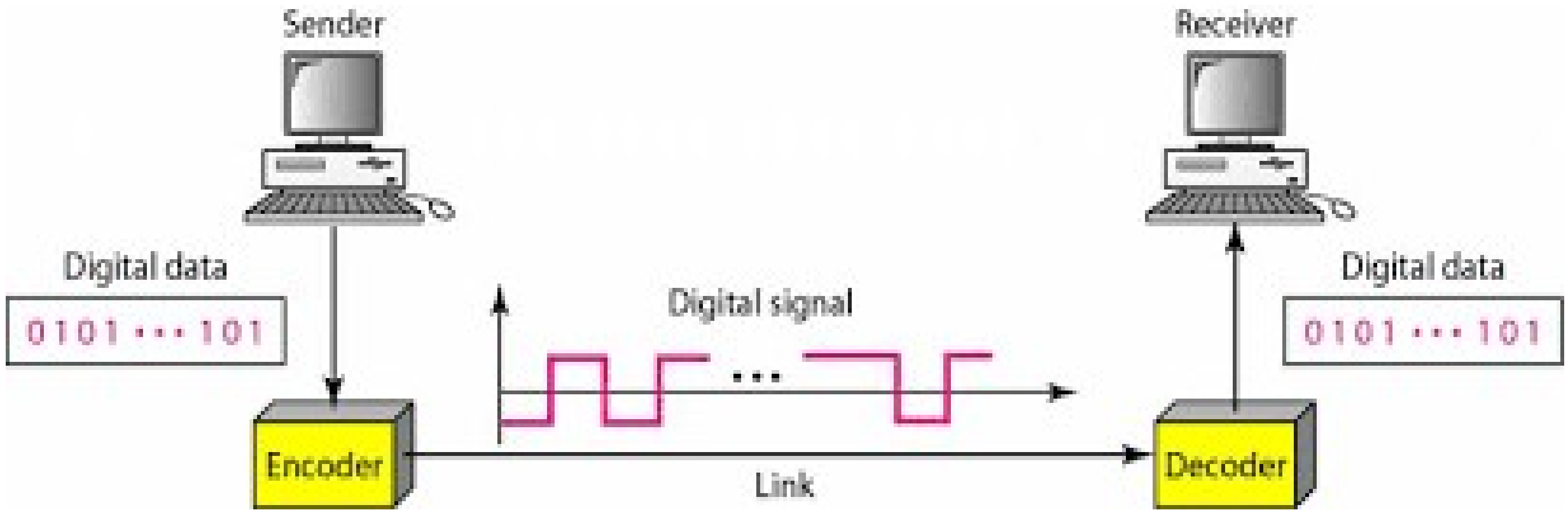
Example: Human voice, Thermometer, Analog phones etc.

- ***Digital Signal:-***Data transmission between computers is in the form of digits (binary digits 0 & 1)

Example : Computers, Digital Phones, Digital pens, etc.



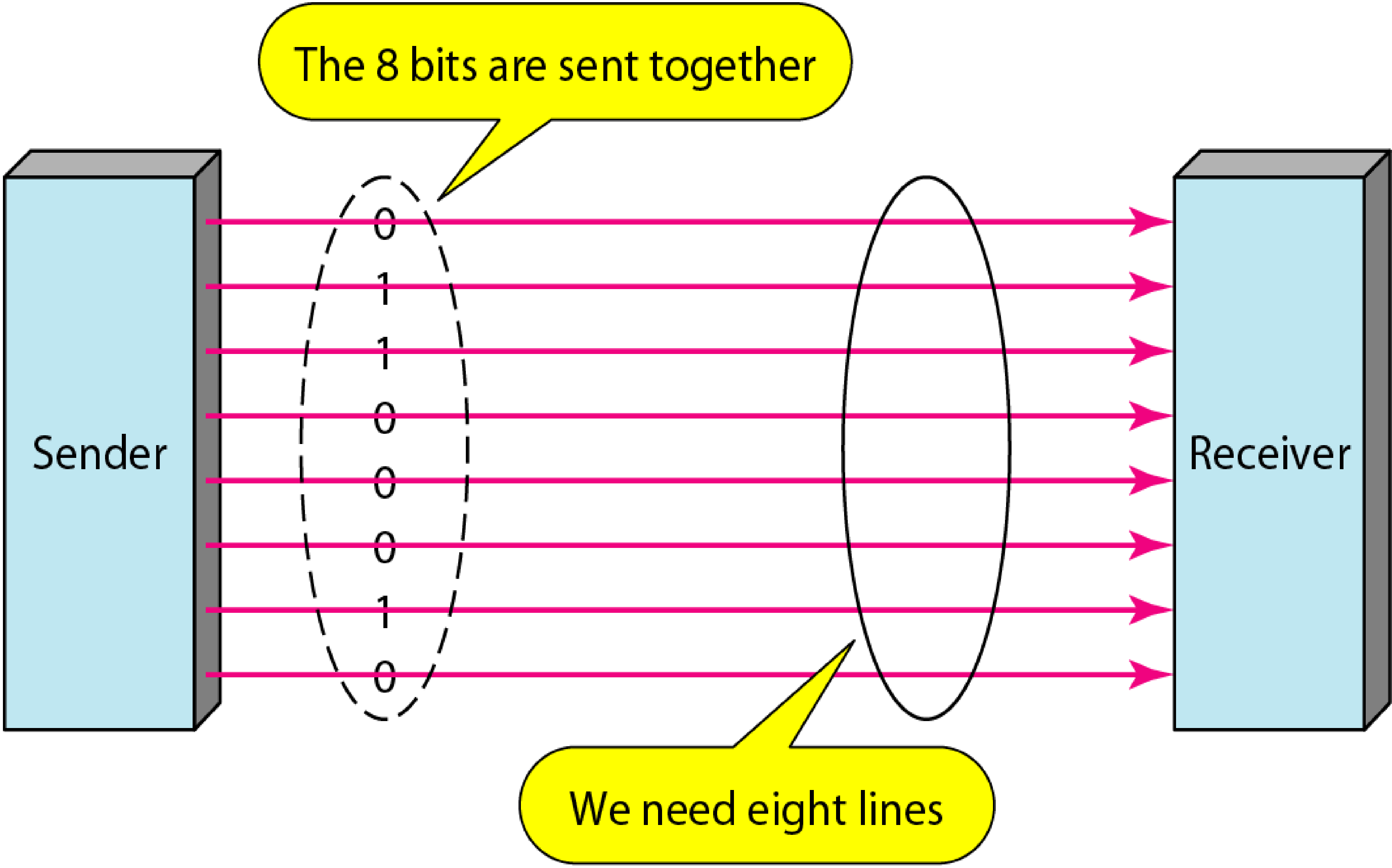
Analog data/ signal to digital data/ signal conversion



Data to signal conversion

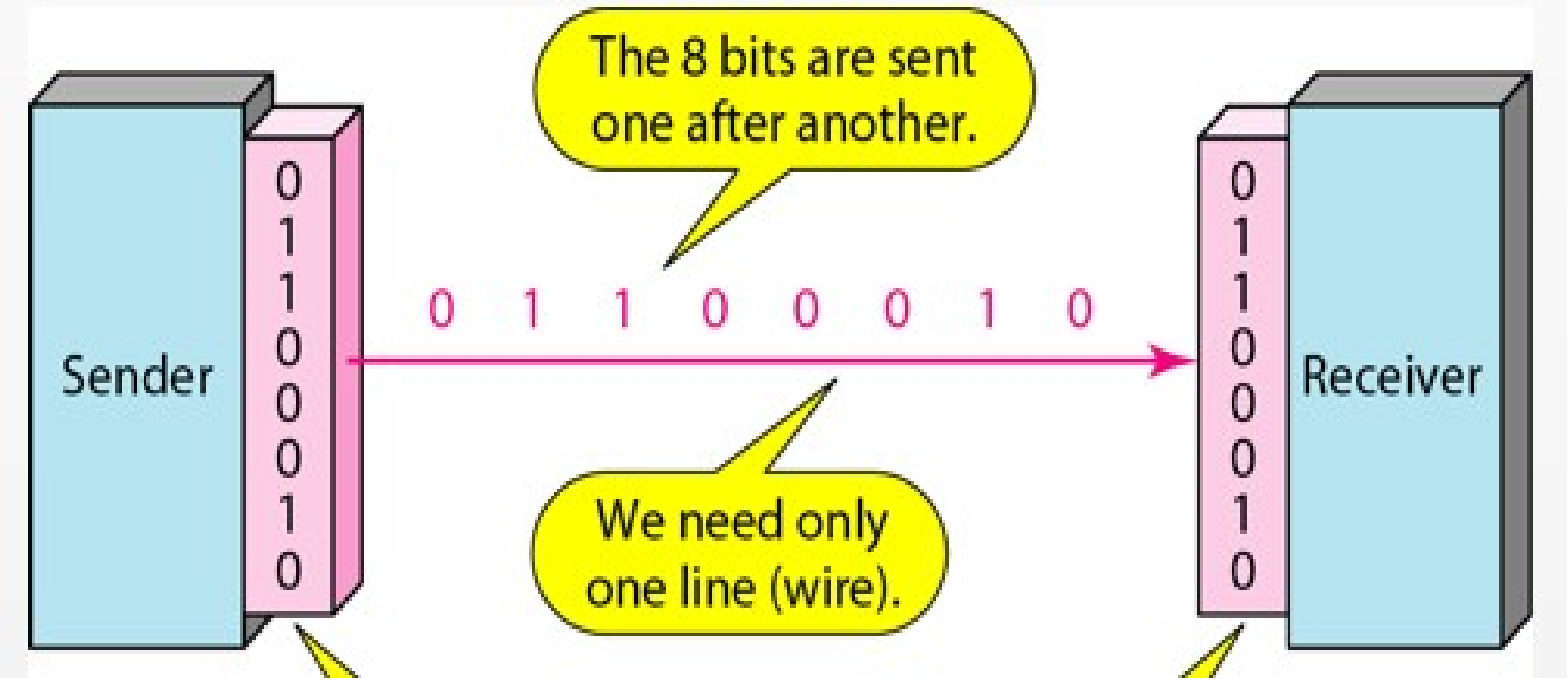
Techniques Of Data Communication

- There are two possible techniques of sending data from the sender to receiver, i.e. parallel and serial transmission.
- **Parallel Transmission**
- Each bit of character / data has a separate channel and all bits of a character are transmitted simultaneously.
- Parallel devices have a wider data bus than serial devices and can therefore transfer data in words of one or more bytes at a time.
- As a result, there is a speedup in parallel transmission bit rate over serial transmission bit rate.


The 8 bits are sent together
Sender
0
1
1
0
0
0
1
0
Receiver
We need eight lines

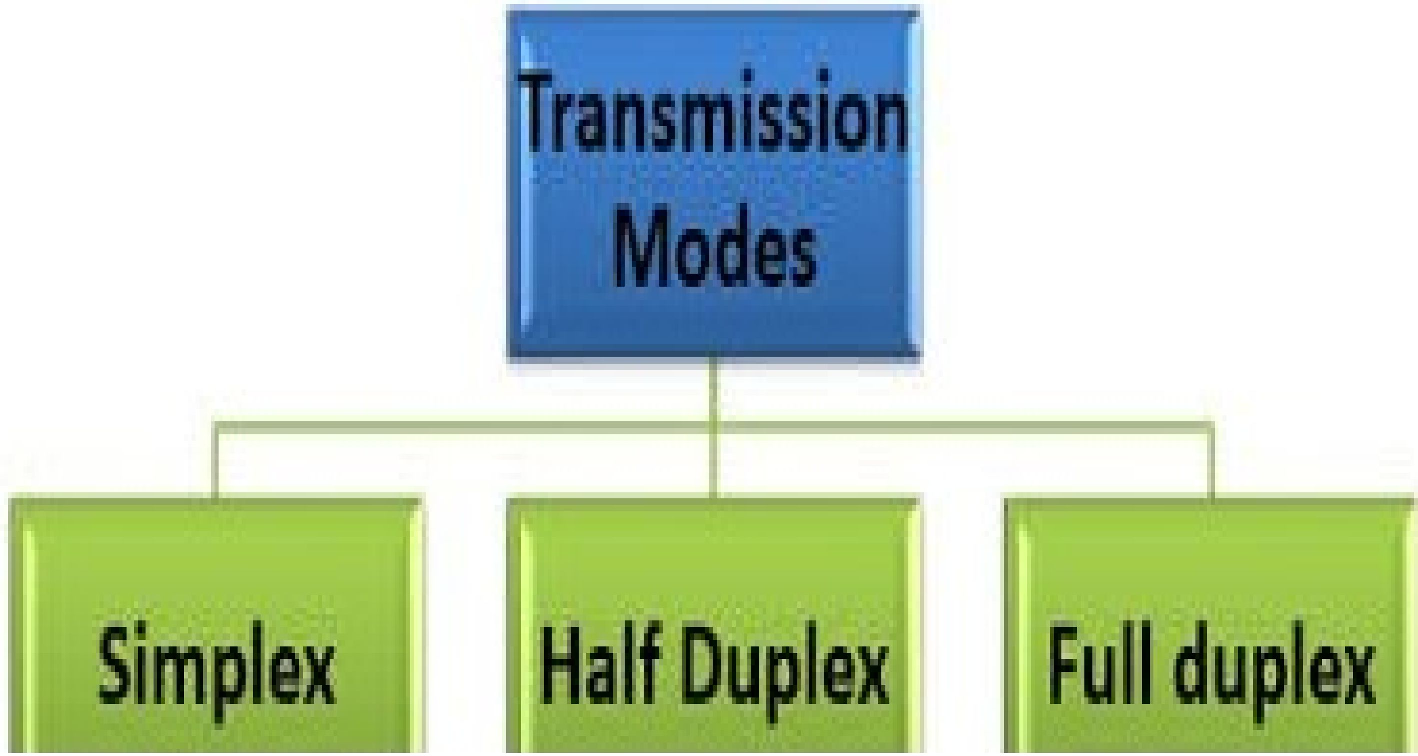
Serial Transmission

- In serial transmission, the data is sent as one bit at a time having a single channel for all the bit.
- Each bit is sent over a single wire, one after the other



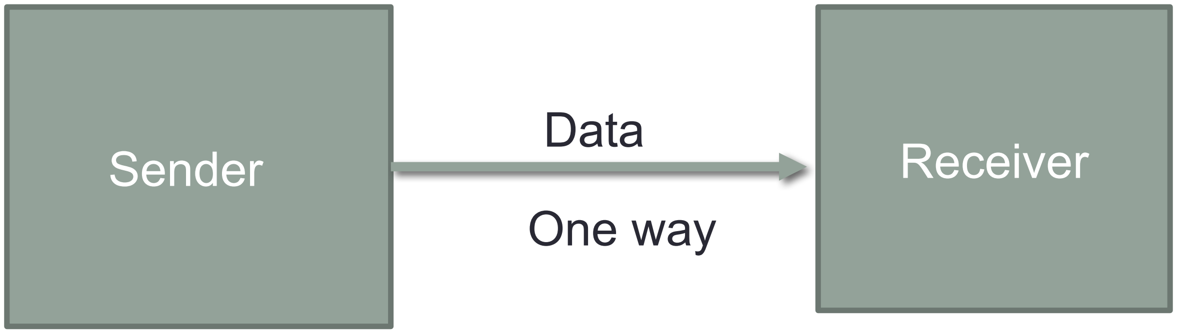
Modes of Data Communication

- There are three ways or modes for transmitting data from one location to another.



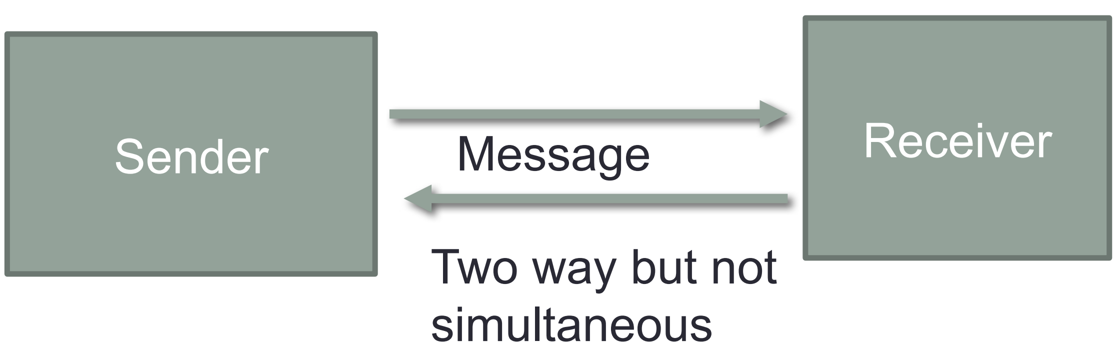
Simplex Mode

- In simplex mode, data is transmitted in only one direction.
- A terminal can only send data and cannot receive it or it can only receive data but cannot send it.
- simplex states that information can only be broadcast in one direction, at one time.



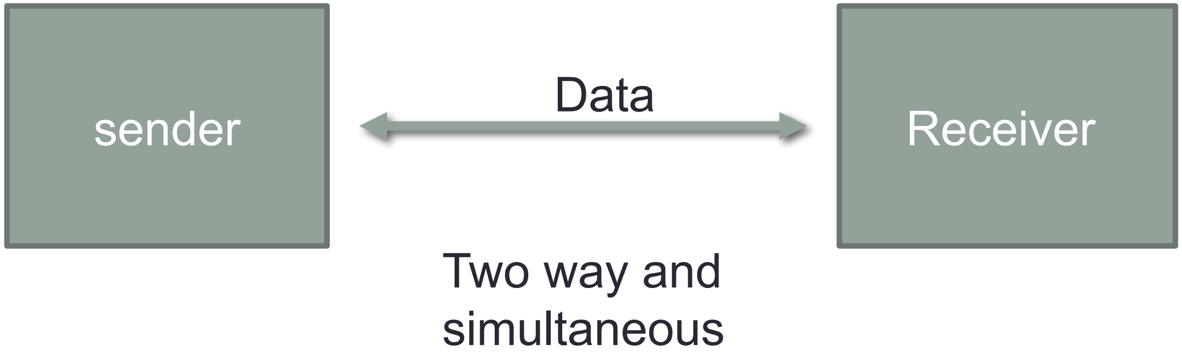
Half Duplex

- In half duplex mode, data can be transmitted in both directions but only in one direction at a time.
- Messages can flow in two directions in a half-duplex type, but never at the same time.
- When one node is transmitting the other can only receiving and vice versa.
- In other words it can be said that at a **single time**, the transmission of data are done in only one direction.



Full Duplex

- In full mode, data can be transmitted in both directions simultaneously.
- It consists of two simplex channels, a forward channel and a backward (reverse) channel, linking at the same points.




TALKING ON THE
PHONE
Street Station

Switching techniques

- These are techniques which are used to establish communication links between data, source and receiver in a communication network.
- There are three forms of switching techniques:-
    - Circuit switching
    - Packet switching
    - Message switching

Circuit Switching

- **In circuit switching**, a connection is established between two network nodes before they begin transmitting data.
- **Circuit switching** is a technique that directly connects the sender and the receiver in an unbroken path.
- With this type of switching technique, once a connection is established, a dedicated path exists between both ends until the connection is terminated.
- A circuit-switching network defines a static path from one point to another.
- As long as the two points are connected, all data traveling between those two points will take the same path.

Phases in Circuit Switching

- **Connection Establishment Phase**
- **Data Transfer Phase**
- **Connection Release Phase**

- E.g. Telephone line

Message switching

- In message switching there is no need to establish a dedicated path between two stations.
- It is also called as Store and forward technology.
- The information is stored and forwarded from the second device after a connection between that device and a third device on the path is established.
- In this technique a node receive a message and store the message until the free and appropriate route is found and if found then send it otherwise stored it.
- Message switching requires that each device in the data's path has sufficient memory and processing power to accept and store the information before passing it to the next node.

Packet Switching

- It can work in both connection oriented and connectionless.
- Packet switching breaks data into packets before they are transported.
- Packets can travel any path on the network to their destination, because each packet contains the destination address and sequencing information.

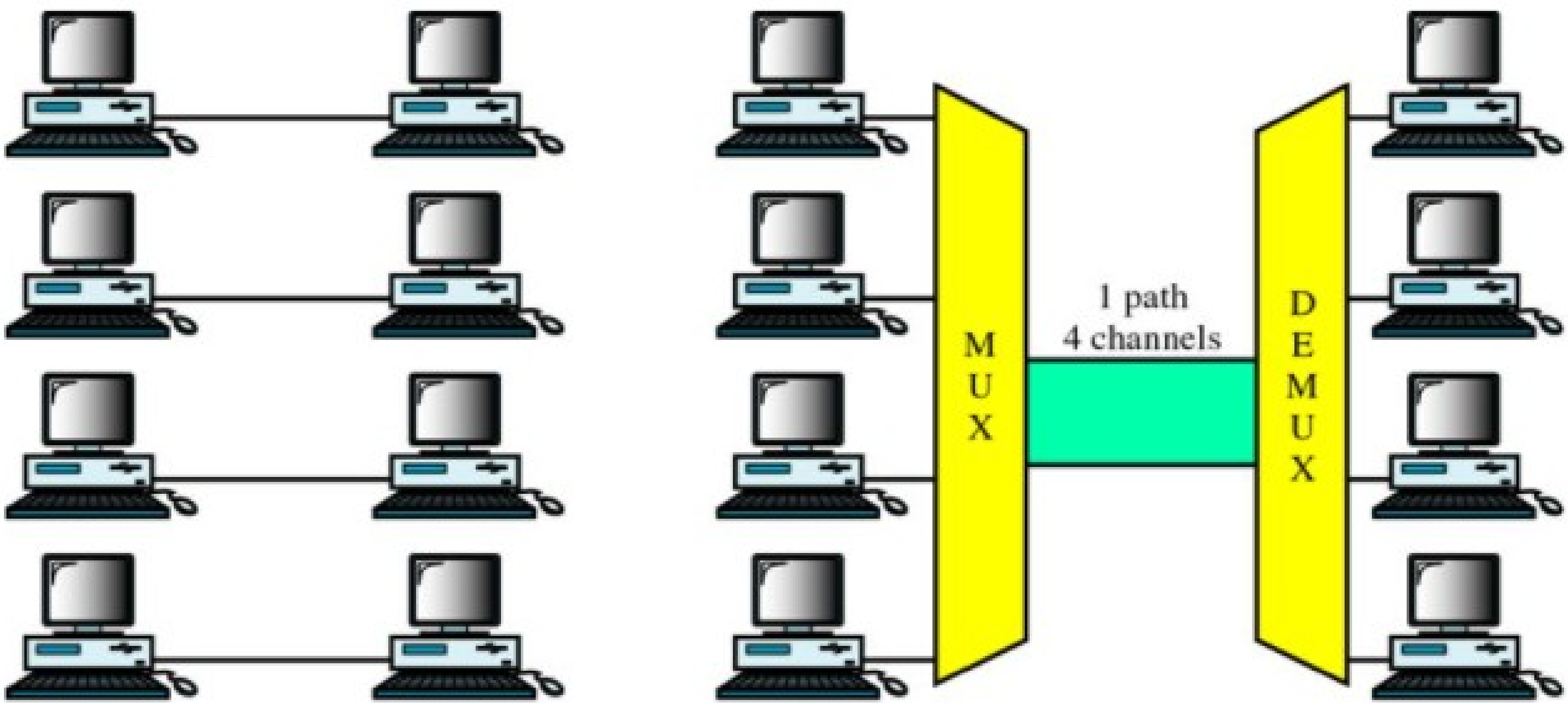
Cont…

- There are three different ways in which packets can be addressed:
  - **Unicast:** packet is addressed to a single destination
  - **Multicast:** packet is addressed simultaneously to multiple destinations
  - **Broadcast:** packet is sent simultaneously to all stations on the network.

Multiplexing and Demultiplexing

- **Multiplexing**
  - sending multiple signals or streams of information over a communications link at the same time in the form of a single and complex signal.
  - the receiver recovers the separate signals, a process called **demultiplexing**.
  - It combine multiple communication signals together in order for them to traverse an otherwise single signal communication medium simultaneously.
  - **Multiplexing** can be applied to both analog and digital signals.
  - A device that performs the multiplexing is called a multiplexer (**MUX**),
  - device that performs the reverse process is called a **demultiplexer** (**DEMUX** or DMX).

Cont…



a. No multiplexing

b. Multiplexing

Transmission Impairments

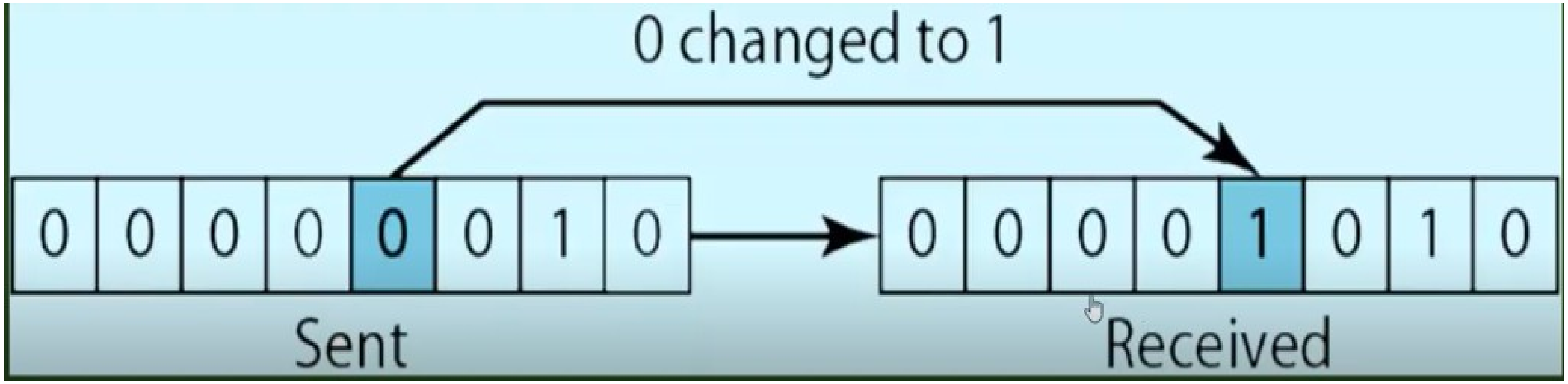
- The signal received may differ from the signal **transmitted**.
- In the data communication system, analog and digital signals go through the **transmission** medium.
- **Transmission** media are **not** ideal. ... So, the signals sent through the **transmission** medium are also **not** perfect. This imperfection cause signal **impairment**.
- There are three types of **transmission impairments**: attenuation, delay distortion, and noise.

Cont…

- **Attenuation**
  - It means loss of energy.
  - The strength of signal decreases with increasing distance which causes loss of energy in overcoming resistance of medium.
  - **Amplifiers** are used to amplify the attenuated signal which gives the original signal back and compensate for this loss.
- **Distortion**
  - It means changes in the form or shape of the signal.
  - This is generally seen in composite signals made up with different frequencies.
- **Noise**
  - The random or unwanted signal that mixes up with the original signal
  - It is the interference of another source of signals

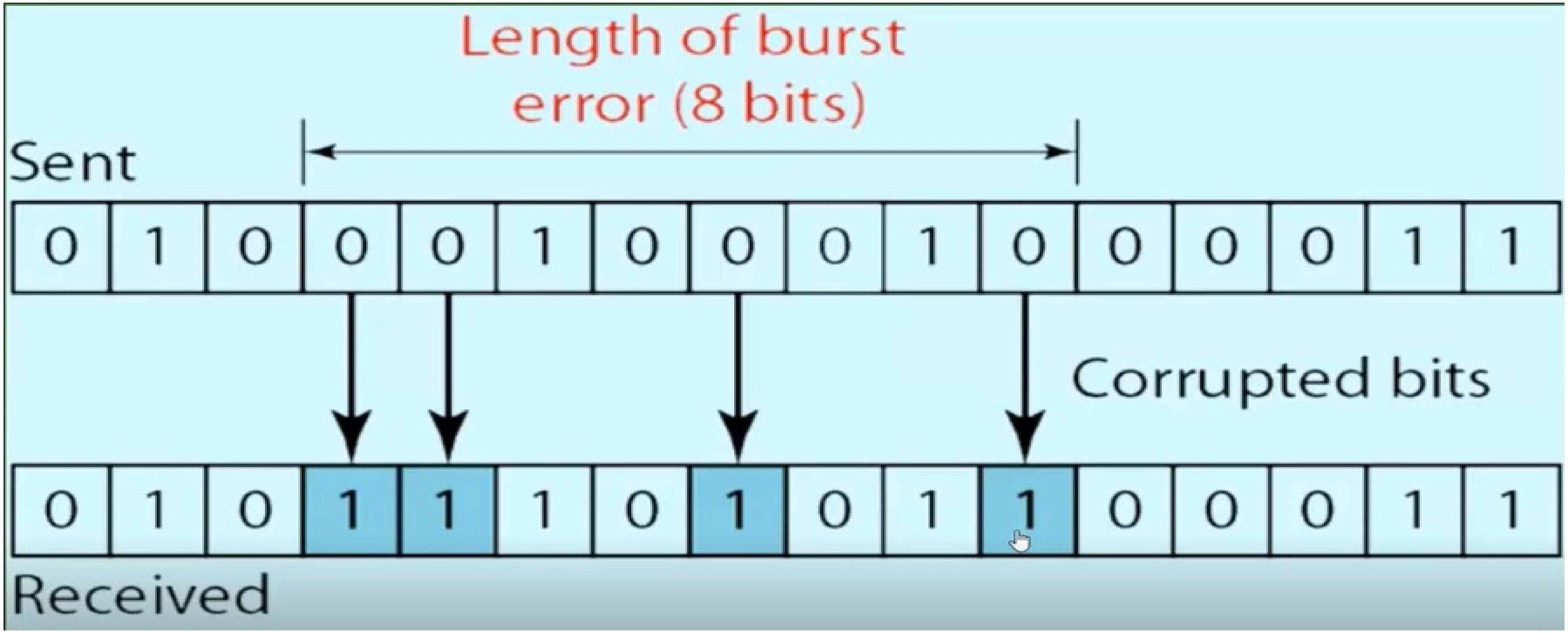
Data Transmission Error Detection and Correction

- Whenever bits flow from one point to another, they are subject to unpredictable changes because of interference.
- This interference can change the shape of the signal.
- In a **single-bit error**, a 0 is changed to a 1 or a 1 to a 0. The term single-bit error means that only 1 bit of a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1.



Cont..

- The term **burst error** means that 2 or more bits in the data unit have changed from 1 to 0 or from 0 to 1.



Cont..

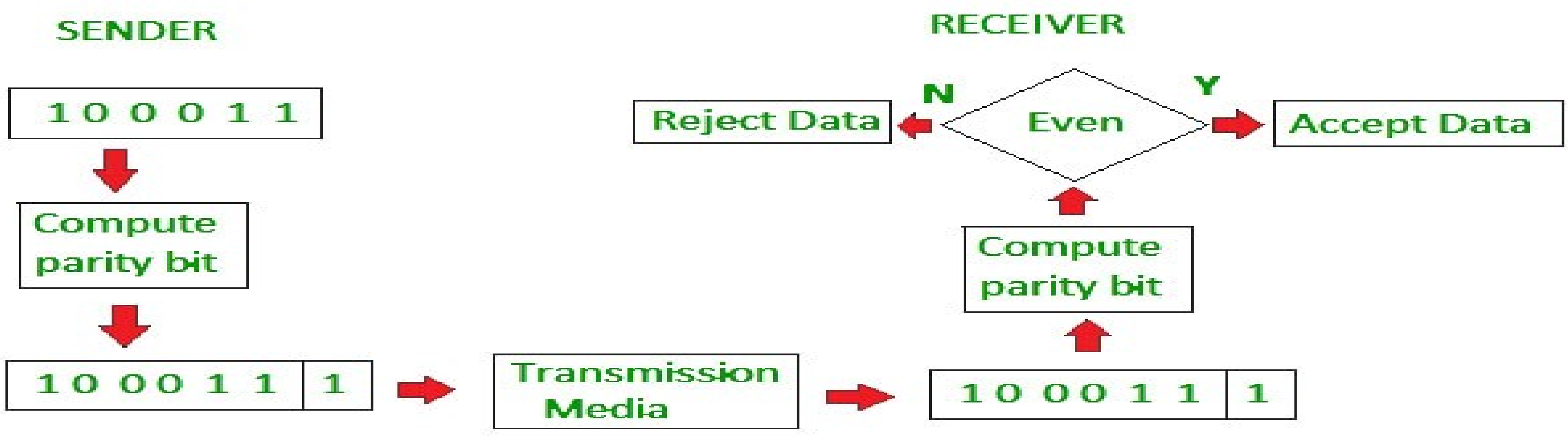
- **Error detection**
  - the **detection** of **errors** caused by noise or other impairments during **transmission** from the transmitter to the receiver.
- **Error correction**
  - the **detection** of **errors** and reconstruction of the original, **error-free data**.
- **Note:** error detection and correction **is not** 100% reliable.
- The most commonly used error detection and correction techniques is **redundancy**.
  - It detects the errors by adding some extra bits to the actual data.

Cont…

- There are 4-types of redundancy techniques

1.Simple Parity Check

- a redundant bit called a parity bit is added to every data unit so that the total number of 1's in the unit (including the parity bit) becomes even (or odd).
- 1 is added to the block if it contains odd number of 1's, and
- 0 is added if it contains even number of 1's
- This scheme makes the total number of 1's even, that is why it is called even parity checking.



Cont…

2. Two-Dimensional Parity Check

- a block of bits is organized in a table (rows and columns).
- First calculate the parity bit for each data unit in both dimensions .
- Then organize them into a table*

Original Data

10011001	11100010	00100100	10000100

	Data	Row parities
	1 0 0 1 1 0 0 1	0
	1 1 1 0 0 0 1 0	0
	0 0 1 0 0 1 0 0	0
	1 0 0 0 0 1 0 0	0
Column parities	1 1 0 1 1 0 1 1	0

100110010	111000100	001001000	100001000	110110110

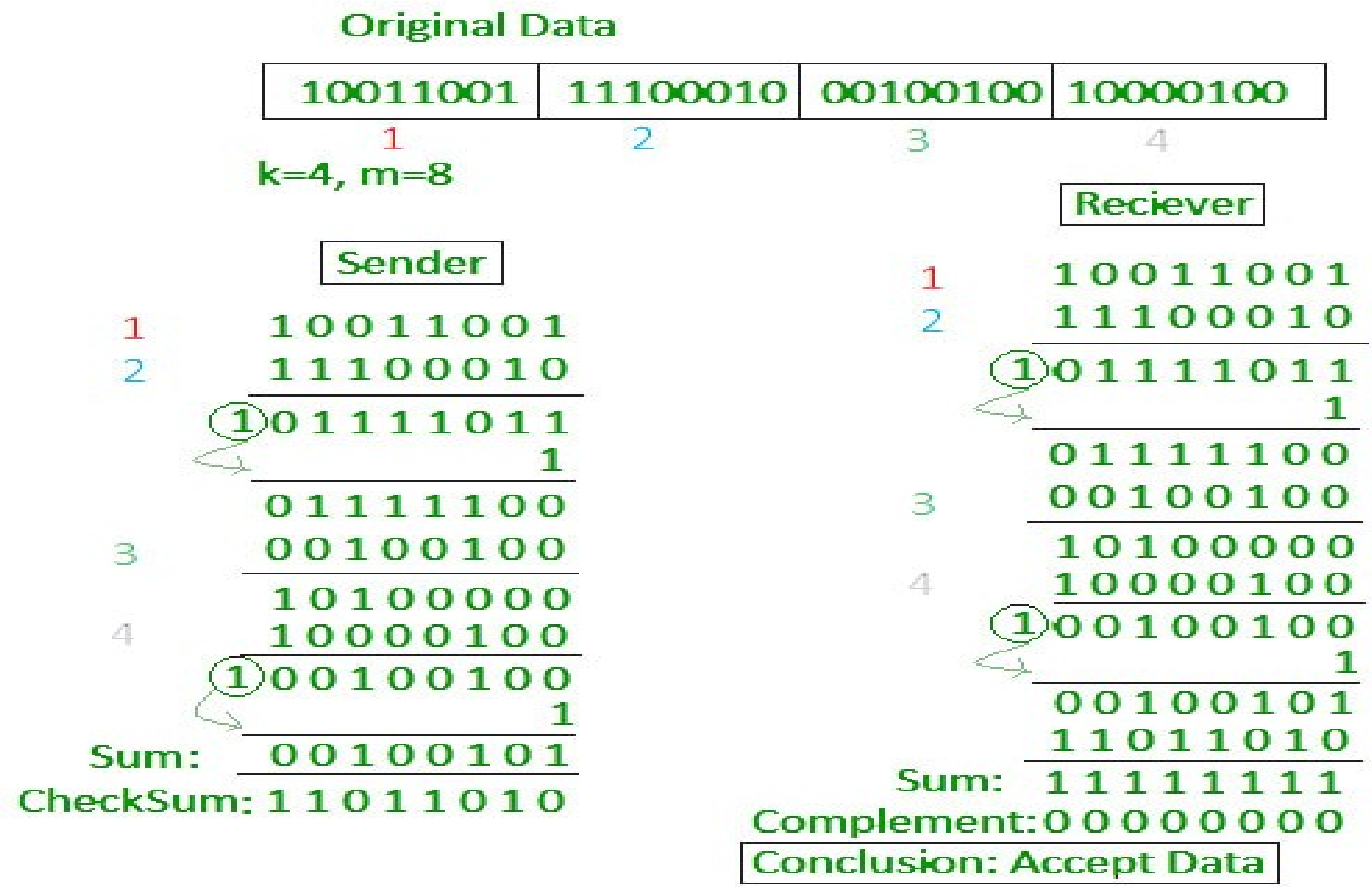
Data to be sent

Cont…

3. Checksum

- The data is divided into k segments each of m bits.
- Then add all the k segments
- Add carry bits if any.
- The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.
- At the receiver's end, all received segments are added including the checksum to get the sum.
- The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.

Cont…


Original Data
10011001
11100010
00100100
10000100
1
2
3
4
k=4, m=8
Sender
1 10011001
2 11100010
1 01111011
1
01111100
3 00100100
10100000
4 10000100
1 00100100
1
Sum: 00100101
CheckSum: 11011010
Reciever
1 10011001
2 11100010
1 01111011
1
01111100
3 00100100
10100000
4 10000100
1 00100100
1
00100101
11011010
Sum: 11111111
Complement: 00000000
Conclusion: Accept Data

Cont…

4. Cyclic redundancy check (CRC)

- Unlike checksum scheme, which is based on addition, CRC is based on binary division.
- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
- At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.

Cont…

original message
1010000

@ means X-OR

Generator polynomial
x^3+1
$1.x^3+0.x^2+0.x^1+1.x^0$
CRC generator
1001 4-bit

If CRC generator is of n bit then append (n-1) zeros in the end of original message

Sender

```
1001 | 1010000000
      @1001
       0011000000
        @1001
         0101000
         @1001
          0011000
           @1001
            01010
            @1001
             0011
```

Message to be transmitted

```
1010000000
      +011
----------
1010000011
```

Receiver

```
1001 | 1010000011
      @1001
       0011000011
        @1001
         01010011
         @1001
          0011011
           @1001
            01001
            @1001
             0000
```

Zero means data is accepted

Cont…

- **Exercise**
  - Check the given data using **two dimensional check** and answer whether the receiver accept the data or not. The data which is delivered to destination are **101011 110101 110000 101110**
  - Generate the checksum and detect the error for the message **1101110111111110000101010**
  - Generate the CRC code at the sender side and detect the error at the receiver side for the message **1000100** using the CRC generator $\mathbf{x^3+X^2+1}$

- Error correction can be handled in two ways
  - **Backward error correction**
    - Once the error is discovered the receiver requests the sender to retransmit the entire data unit
  - **Forwarded error correction**
    - The receiver uses the error correcting code which automatically corrects the errors.

Cont…

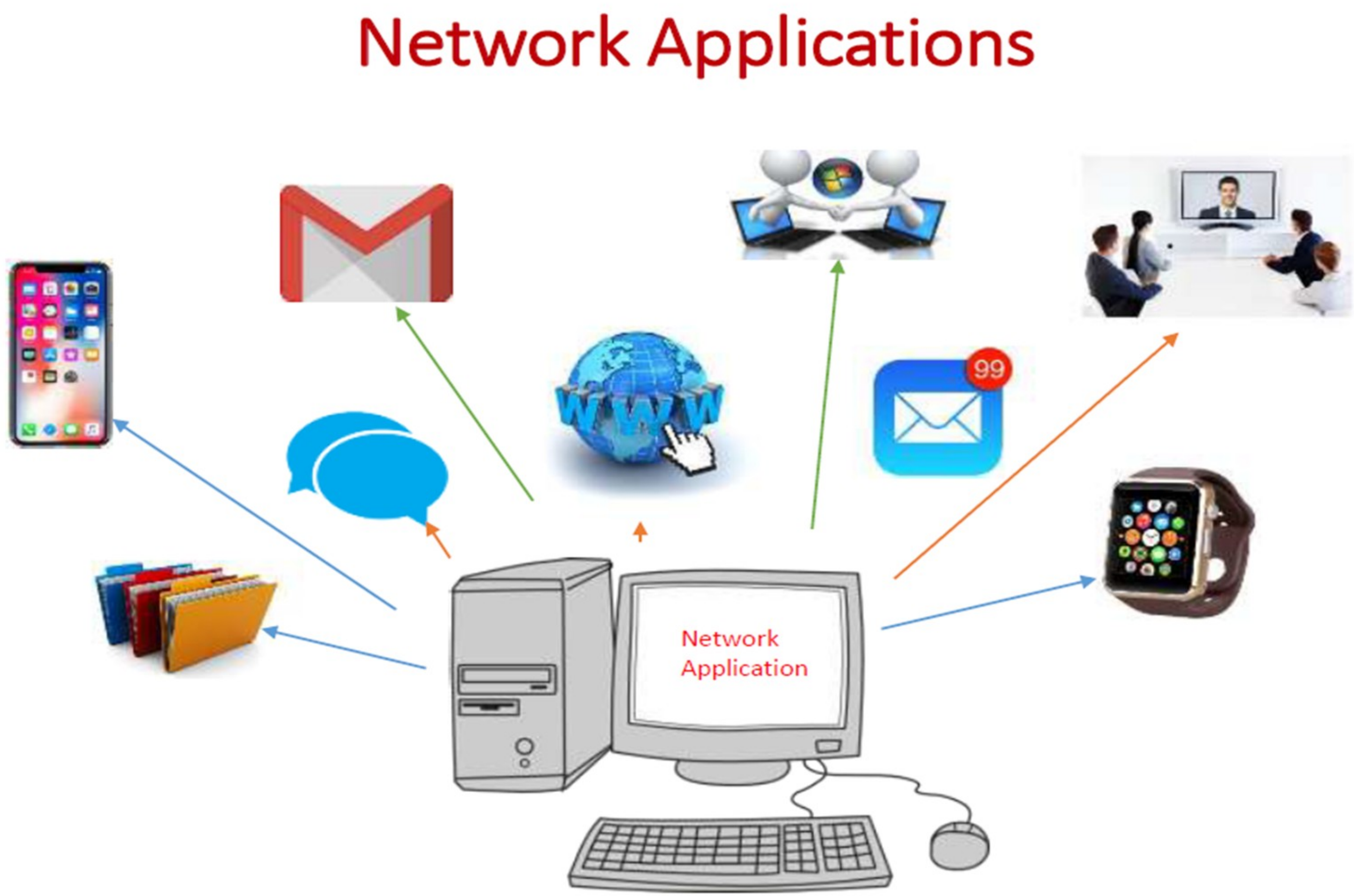
- Home study
  - Types of multiplexing
  - How the detected data is corrected?
    - Hamming code (forwarded)

Computer network

Computer network

- A computer network is an interconnection of two or more computers that are able to exchange information.
- The computer may be connected via any data communication link, like copper wires, radio links, etc.
- They may be personal computers or large main frames.
- The computer network may be located in a room, building, city, country, or anywhere in the world.

Applications of network



Network Criteria

- networks form the backbone of communication, data transfer, and computational power across systems.
- Whether for small organizations or large global corporations, understanding network criteria is essential to ensure efficient, secure, and resilient networks.
- Network criteria define the parameters and standards that networks must meet to perform effectively, which include aspects such as performance, reliability, scalability, and security.
- In this chapter, we will delve into each of these criteria, exploring the standards and configurations necessary to design and maintain robust networks.

Network Criteria

Performance

- Performance in a network refers to the effectiveness of data transfer between devices within the network.
- This criterion is often evaluated by considering factors like throughput and latency.

Reliability

- Network reliability is crucial for ensuring continuous operation without interruptions.
- It measures how often a network remains operational and its capacity to maintain connections and data flow without failure

Network Criteria

Scalability

- Scalability is the ability of a network to grow and expand efficiently without performance loss.
- networks need to accommodate more devices, users, and data.
- Scalable networks are designed to be flexible, allowing for easy integration of new resources.

Security

- Network security is a primary criterion, focusing on protecting data and systems from unauthorized access, attacks, and data breaches.
- Security mechanisms include firewalls, intrusion detection systems, encryption, and multi-factor authentication.

Network Criteria

Bandwidth

- Bandwidth is the maximum amount of data that can be transferred across a network at a given time.
- High bandwidth ensures that large amounts of data can be transmitted efficiently, reducing the likelihood of bottlenecks. Bandwidth requirements vary depending on the application;

Availability

- Network availability is a measure of the percentage of time a network is operational and accessible to users.

Network Criteria

Fault Tolerance

- Fault tolerance is a network's ability to continue operating in the event of a component failure.

Throughput

- Throughput represents the actual rate at which data is successfully transmitted through a network.

Quality of Service (QoS)

- QoS refers to managing network resources to ensure a certain level of performance for different applications and users.

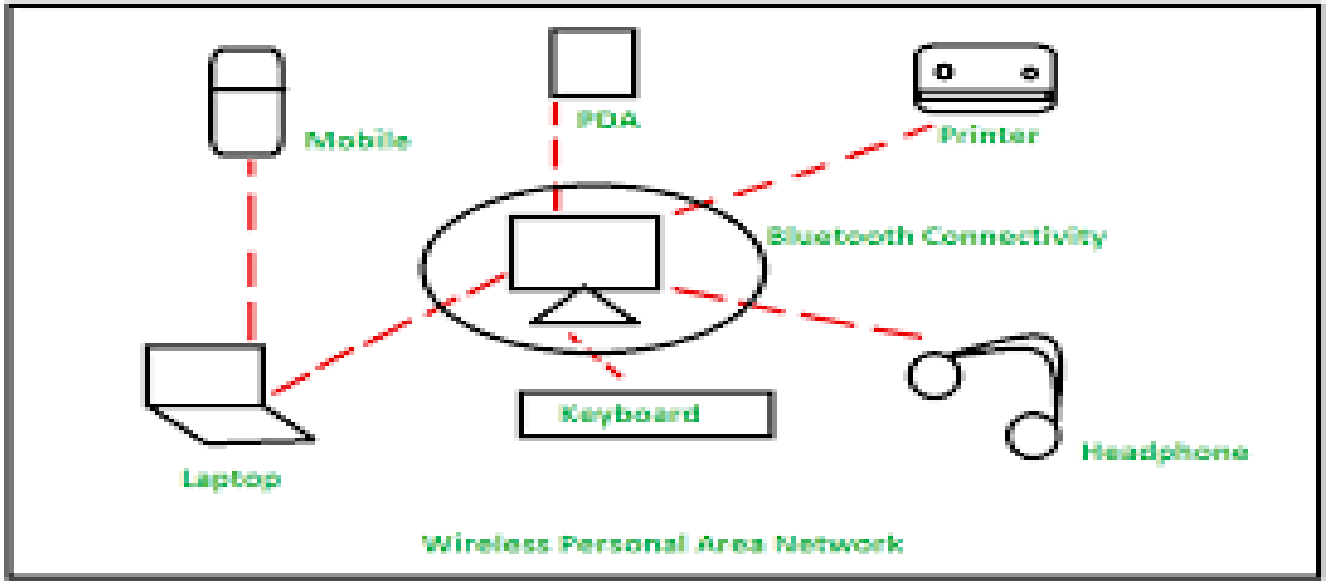
Types of network

Types of network

- Networks are categorized based on the **geographical area they cover** and **their architecture**.
- Based on the geographical area they cover
  - PAN
  - LAN
  - MAN
  - WAN
- Based on their architecture
  - Peer-to-peer
  - Client-server

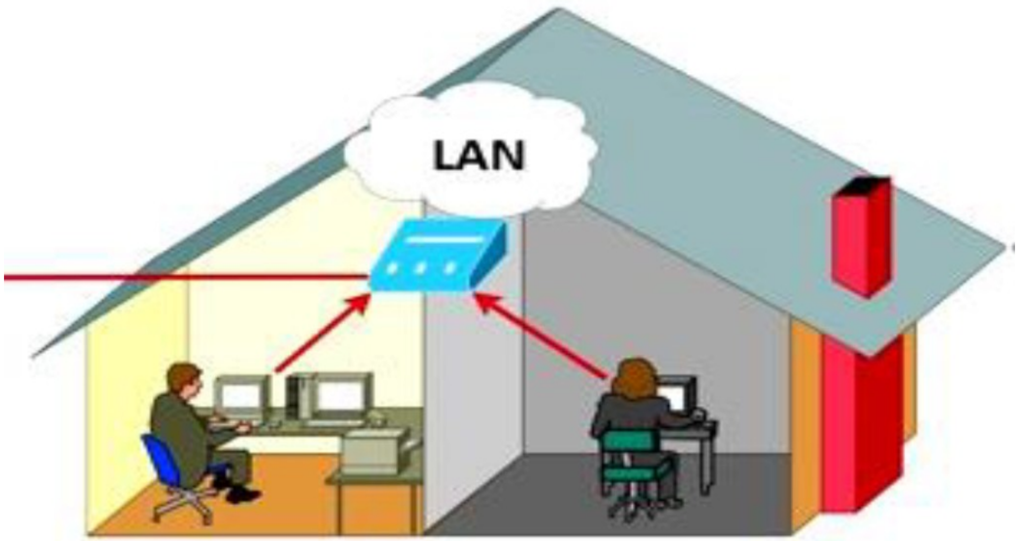
Based on geographical area they cover

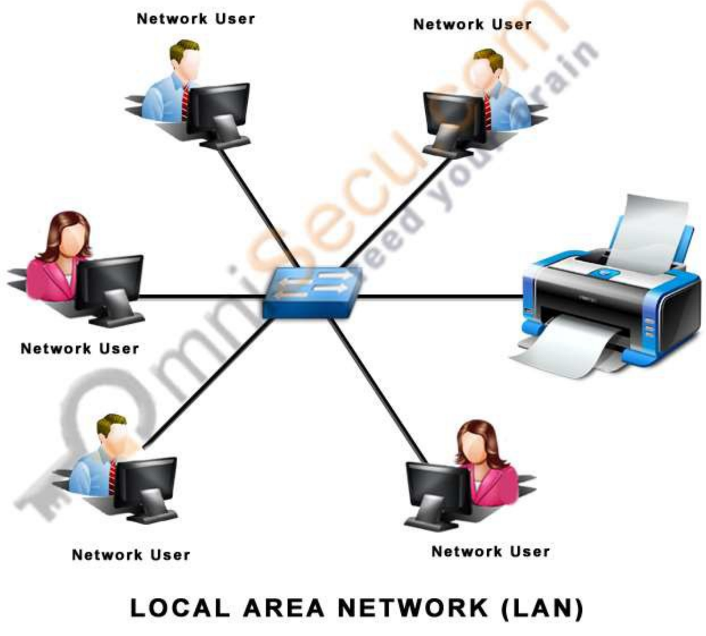
- PAN (Personal Area Network)
- PAN is a network for interconnecting electronic devices within an individual person's workplace.
- It provides data transmission among devices such as computers, smartphones, tablets.



LAN (Local Area Networks)

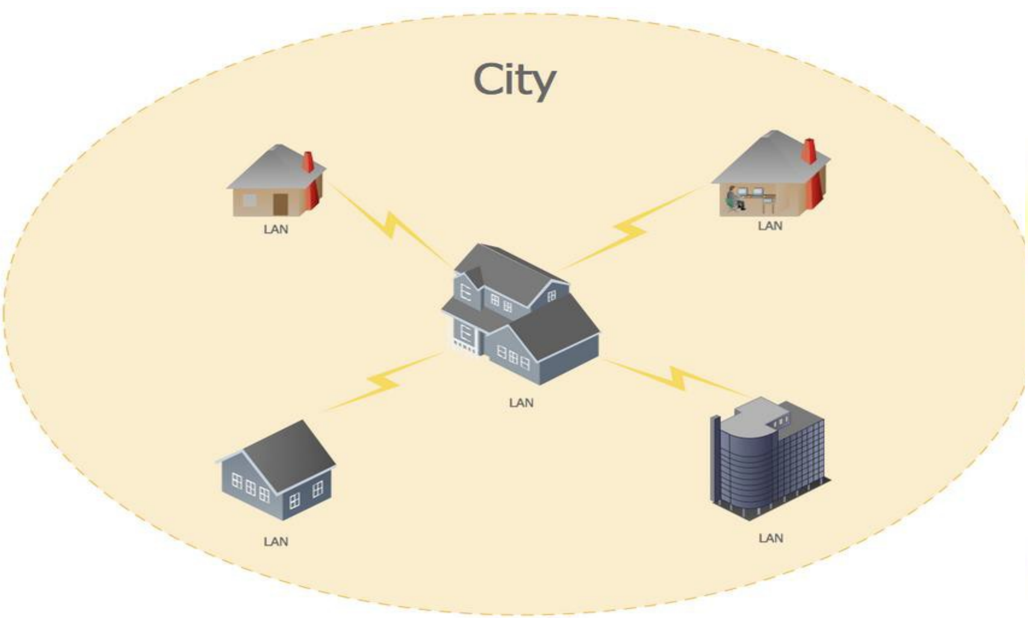
- LAN is a computer network widely used for local communication.
- LAN connects computers in a small area like a room, office, building or a campus spread up to a few kilometers.
- They are privately owned networks, to exchange information.





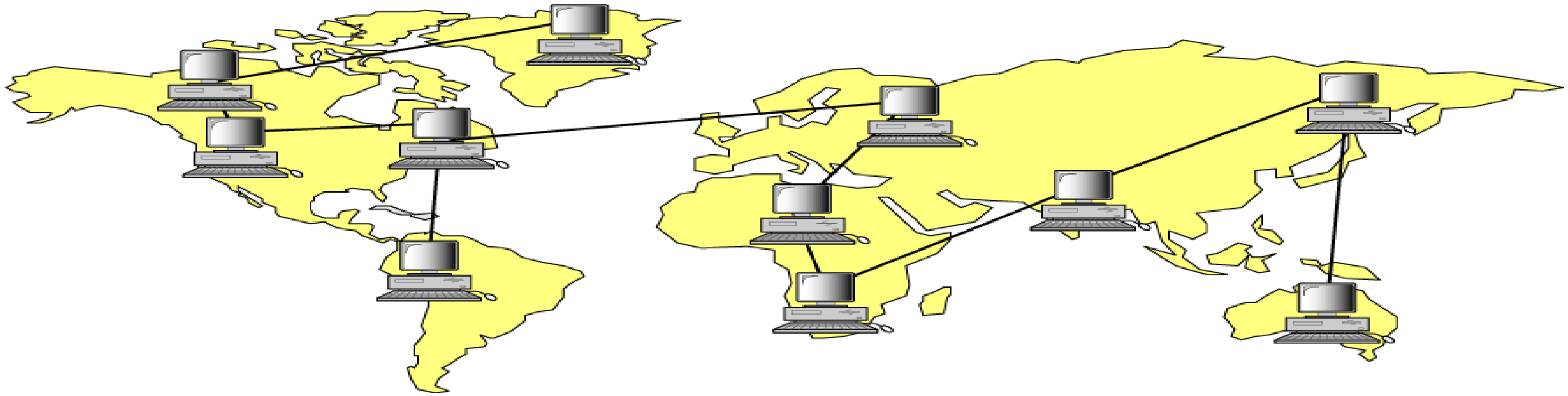
MAN (Metropolitan Area Network)

- MAN is a computer network spread over a city.
- The computers in a MAN are connected using cables.
- MAN connects several LAN spread over a city.
- Example- connection between branches



WAN (Wide Area Network)

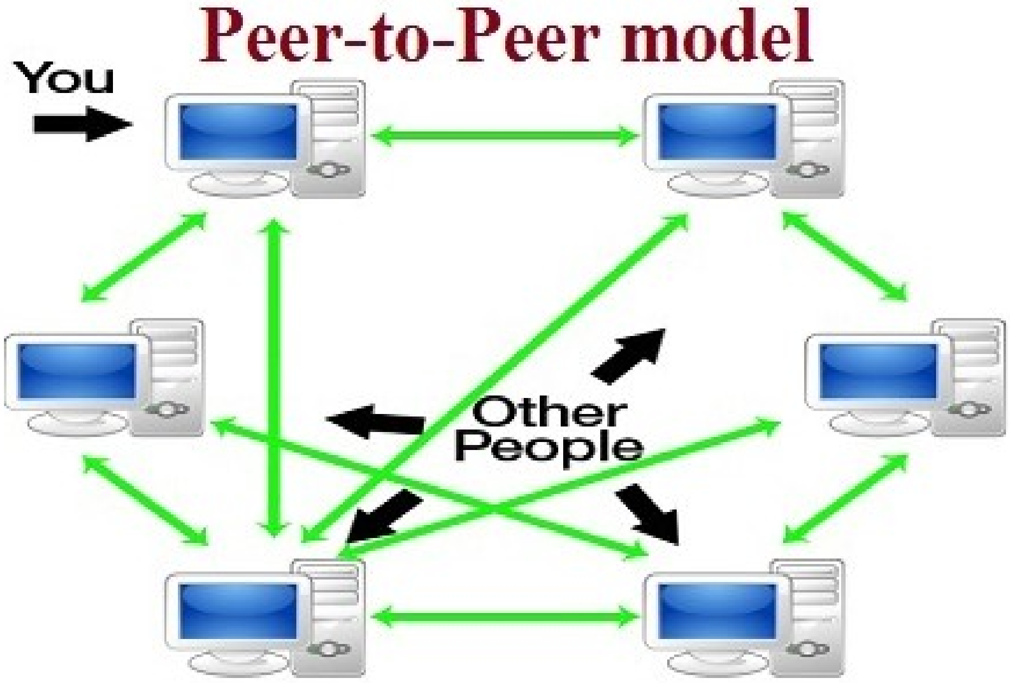
- WAN is a network that connects computers over long distances like cities, countries, continents or world wide.
- WAN uses public, leased, or private communication links to spread over long distances.
- WAN uses telephone lines, satellite link and radio link to connect.
- Using INTERNET in public network is a common example of WAN.



Based on their architecture

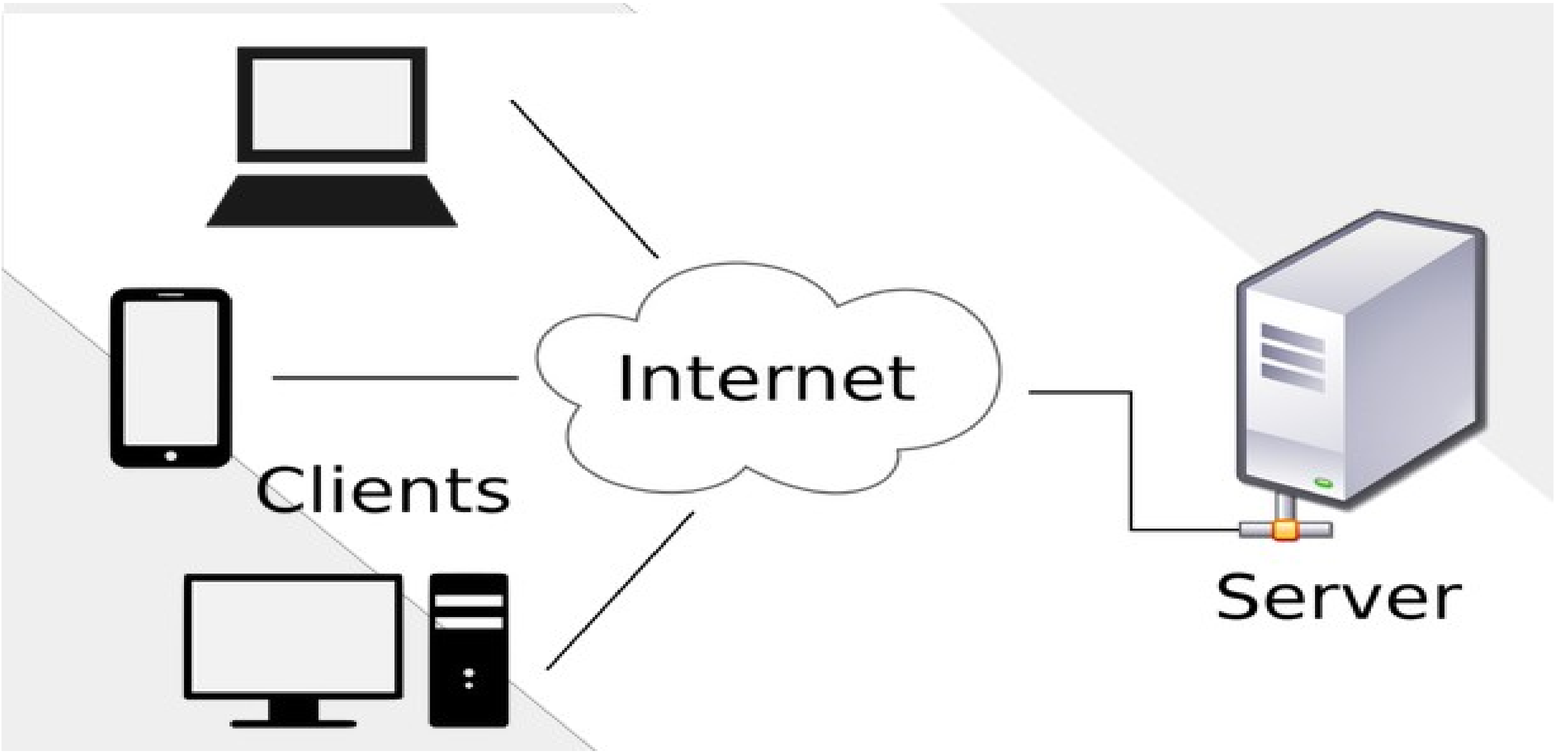
Peer to peer network

- Two or more PCs share files and access to devices such as printers without requiring a separate server computer or server software.



Client-server network

- Where client request for a service to a server and server replies according to client request.



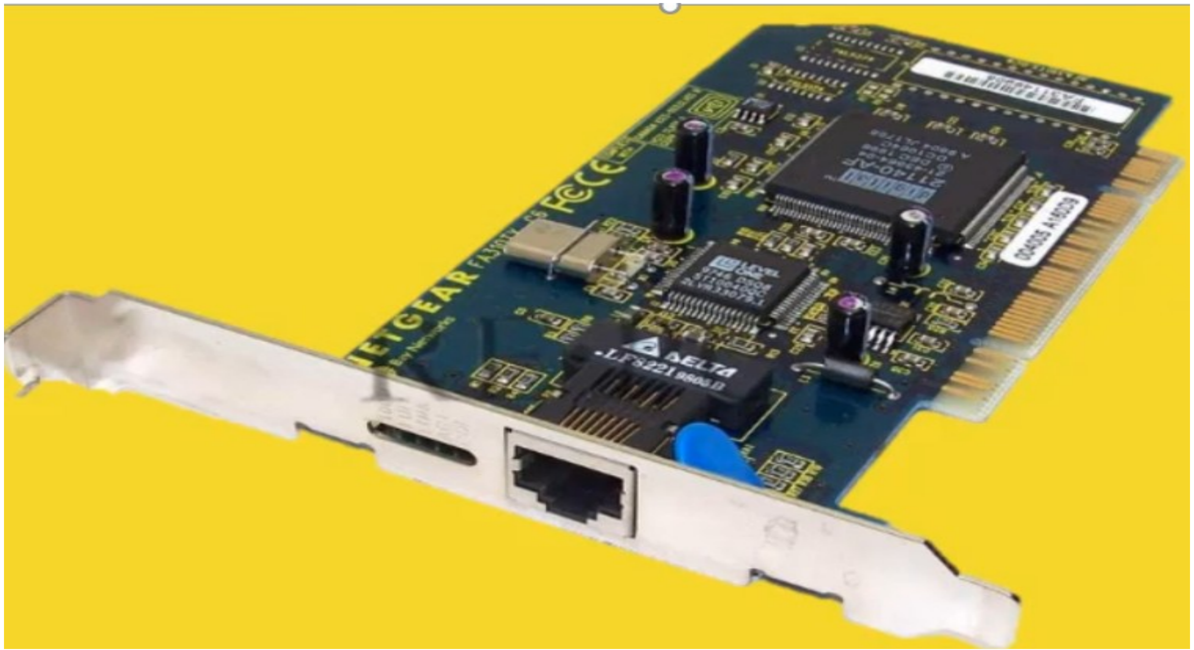
Network Components

Network Components

- Computer networks contains both physical or hardware and software components.
- The software components are **operating systems** and **protocols**.
- The hardware components are the **server**, **client**, **peer**, **transmission medium** and **connecting devices**.
- **Network devices** are the devices that interconnect networks. Because these devices connect network entities they are known as **connecting devices.** These devices include:
    - NIC
    - Hub
    - Repeater
    - Switch
    - Bridge
    - Router
    - Gateway

NIC/ network interface card

- NIC provides the physical interface between computer and cabling.
- It prepares data, sends data, and controls the flow of data. It can also receive and translate data into bytes for the CPU to understand.



Hub

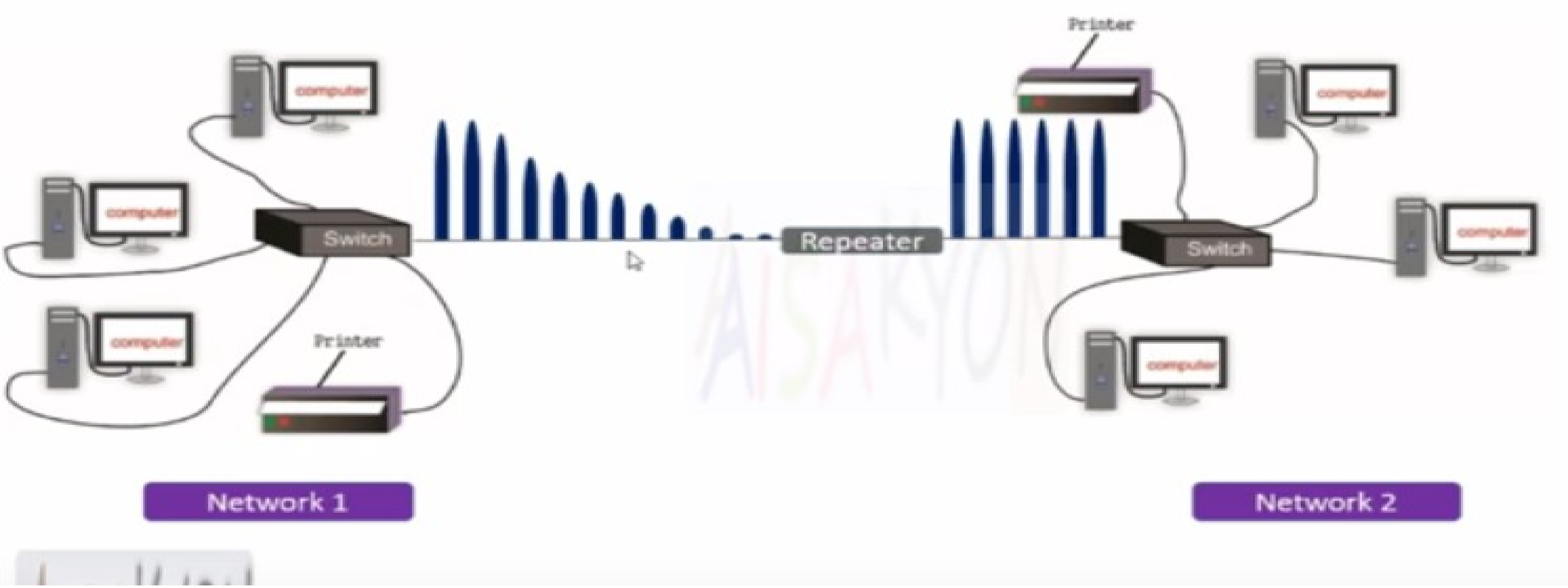
- A hub is a multiport connecting device that is used to interconnect LAN devices.
- A hub can be used to extend the physical length of a network.
- Note: A hub just repeats all the data received on any port to all the other ports; thus, **hubs are also known as repeaters.**

Switch

- It interconnects different devices for sharing data
- It receives incoming data packets and redirects them to their destination on a local area network (LAN).

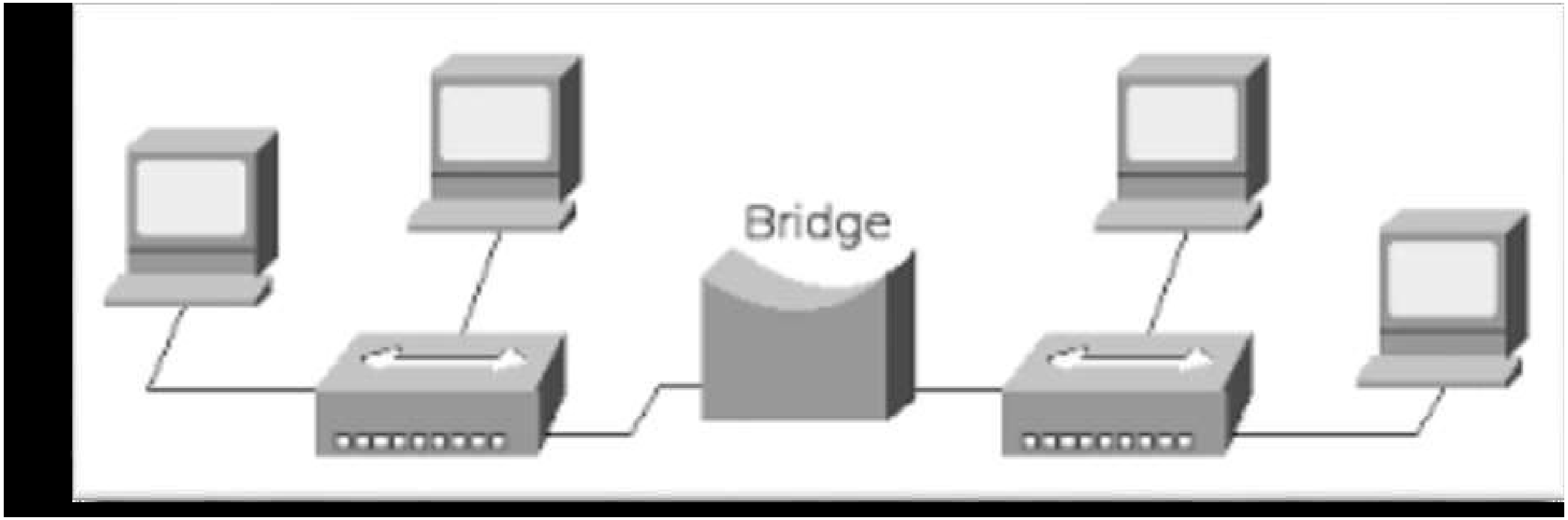
Repeater

- Repeater boost or amplifies the signal before passing it through to the next section of cable.
- used to an extended geographical or topological network boundary



Bridges

- It connects the network with same protocol and topology.
- The main task of a bridge computer is to receive and pass data from one LAN to another.

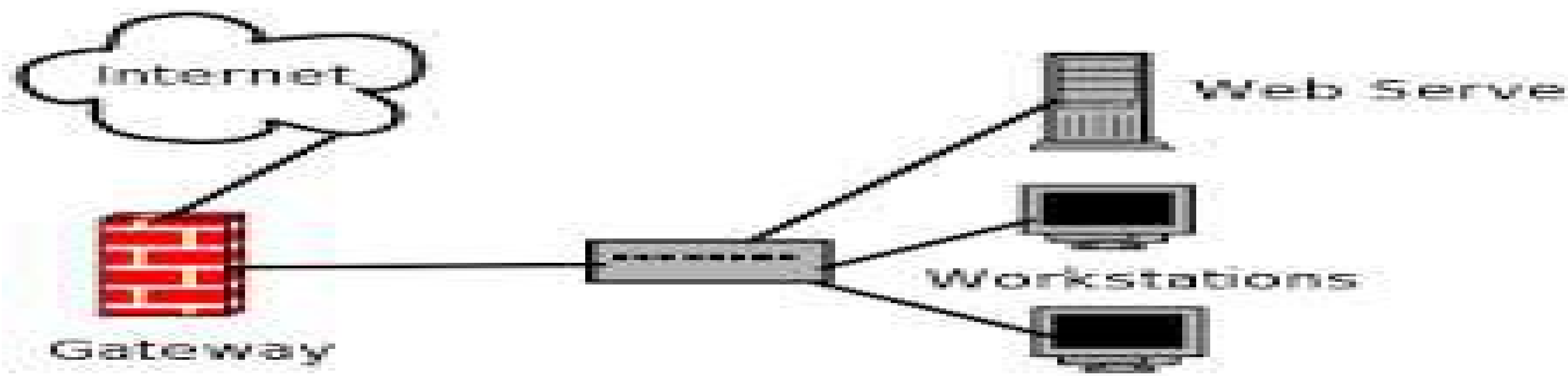


Routers

- A router is a device that connects multiple networks using similar or different protocols.
- Routers are used when several networks are connected together.

Gateway

- Gateway is a device that connects two or more networks with different types of protocol.
- It receives data from one network and converts it according to the protocol of other network.



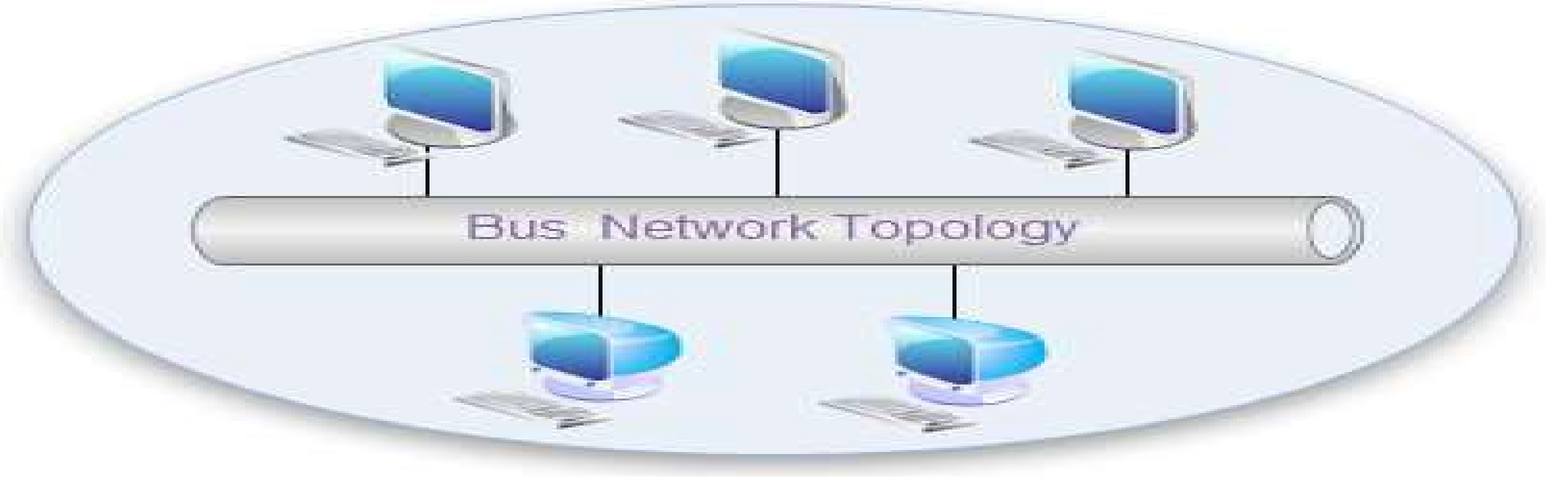
Network Topology

Network Topology

- It is the physical way in which computers are interconnected.
- Five basic network physical structures are :
  - Bus topology
  - Ring topology
  - Star topology
  - Mesh topology
  - Hybrid topology

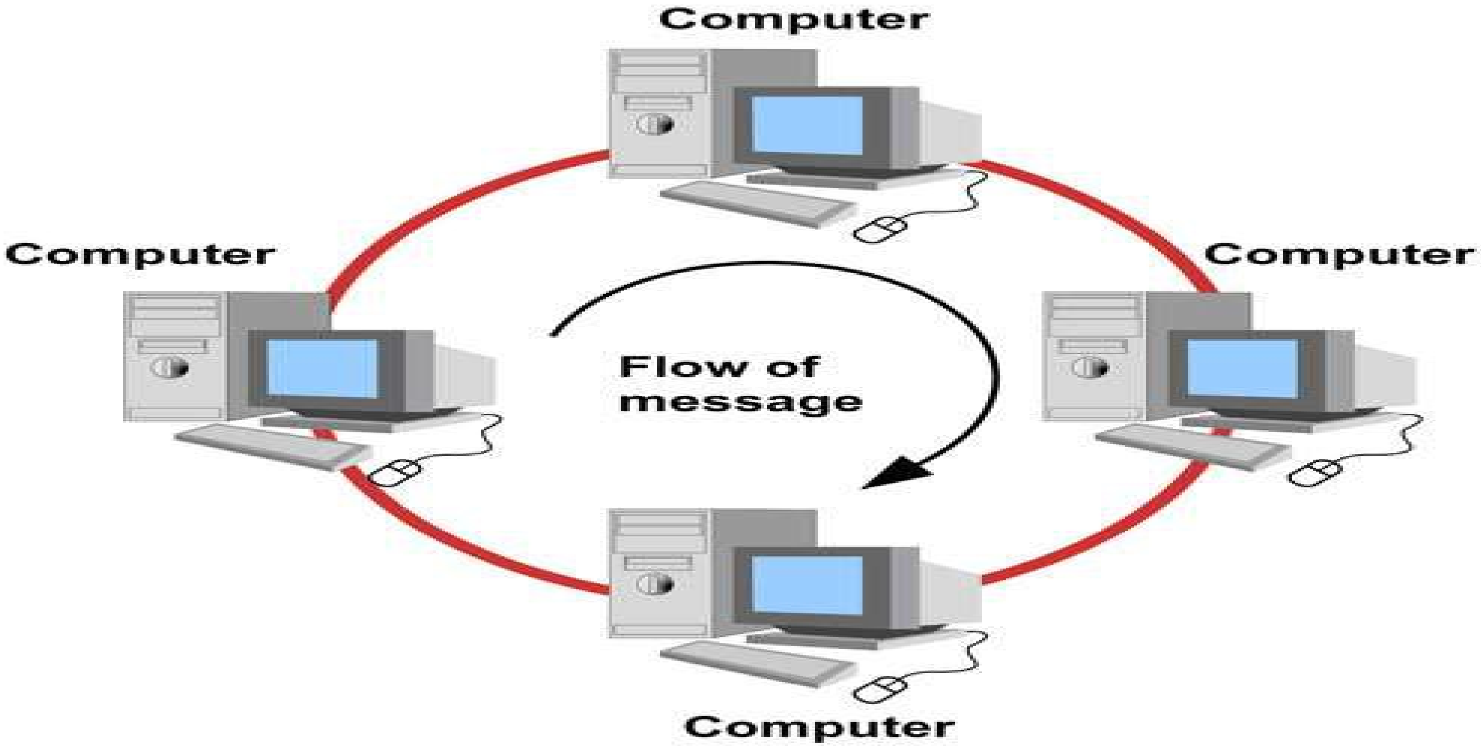
Bus topology

- It consists of a single cable called a trunk (also called a backbone or segment) that connects all of the computers in the network in a single line.
- In Bus topology a single network cable runs in the building or campus and all nodes are linked along with this communication line with two endpoints called the bus or backbone.
- This structure is very popular for local area networks



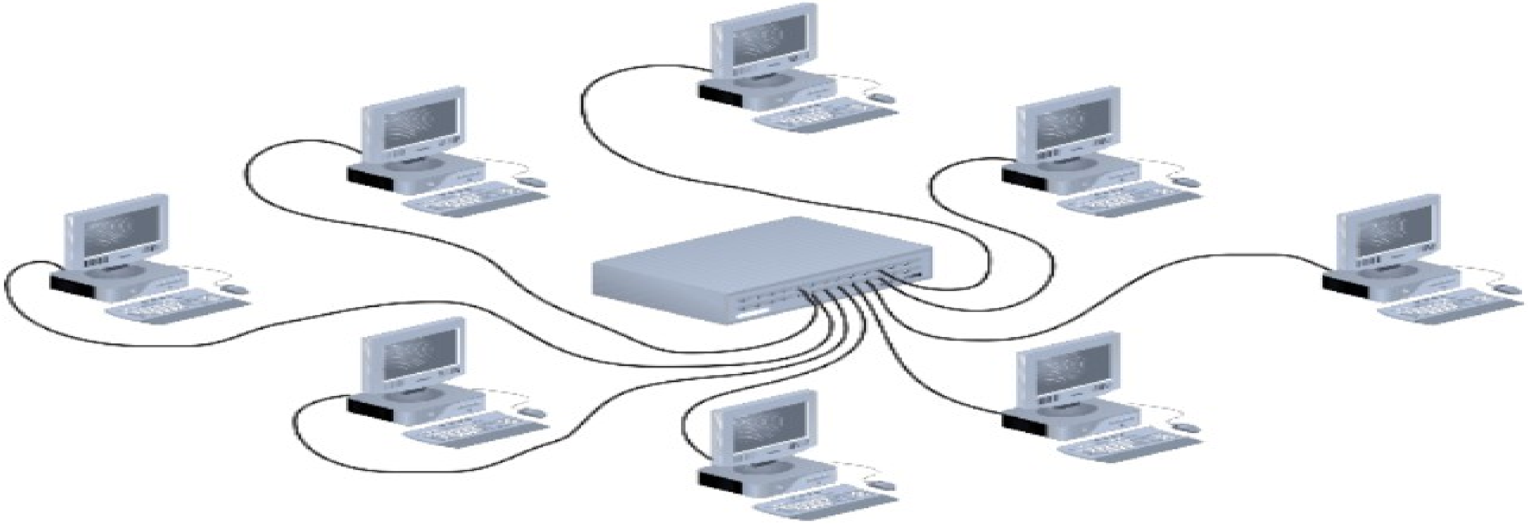
Ring topology

- In Ring topology the network cable passes from one node to another until all nodes are connected in the form of a "loop or ring".
- Transmits in only one direction.



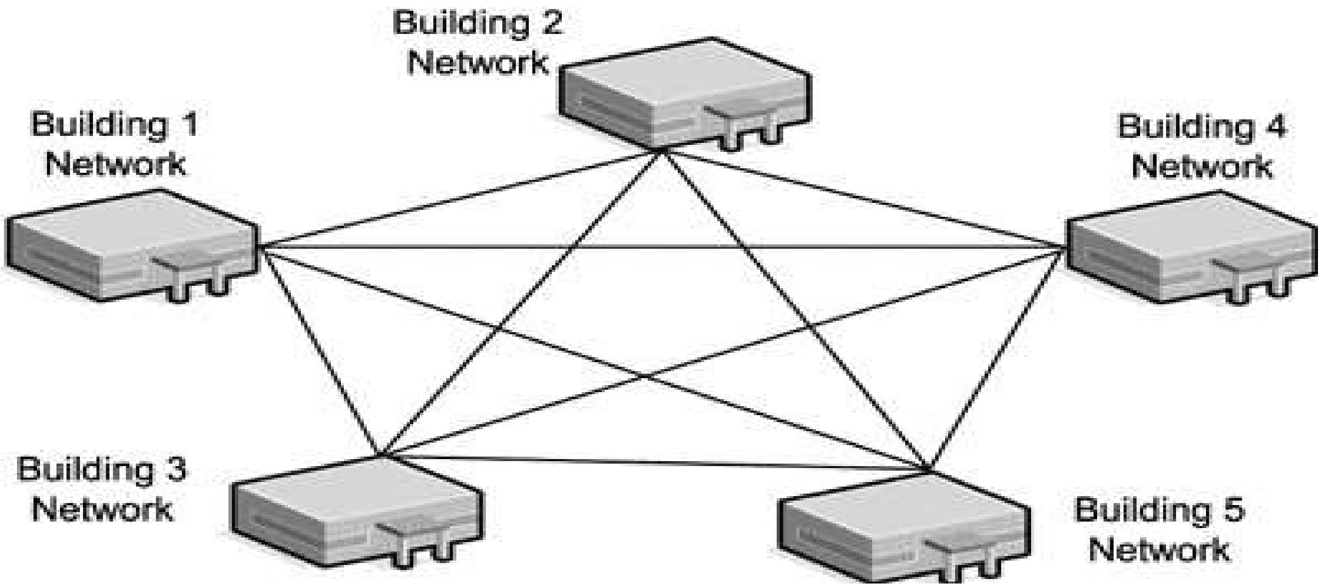
Star topology

- Devices are connected to a central device
- Signals are transmitted from the sending computer through the hub to all computers on the network.
- Because each computer is connected to a central point, this topology requires a great deal of cable in a large network installation.



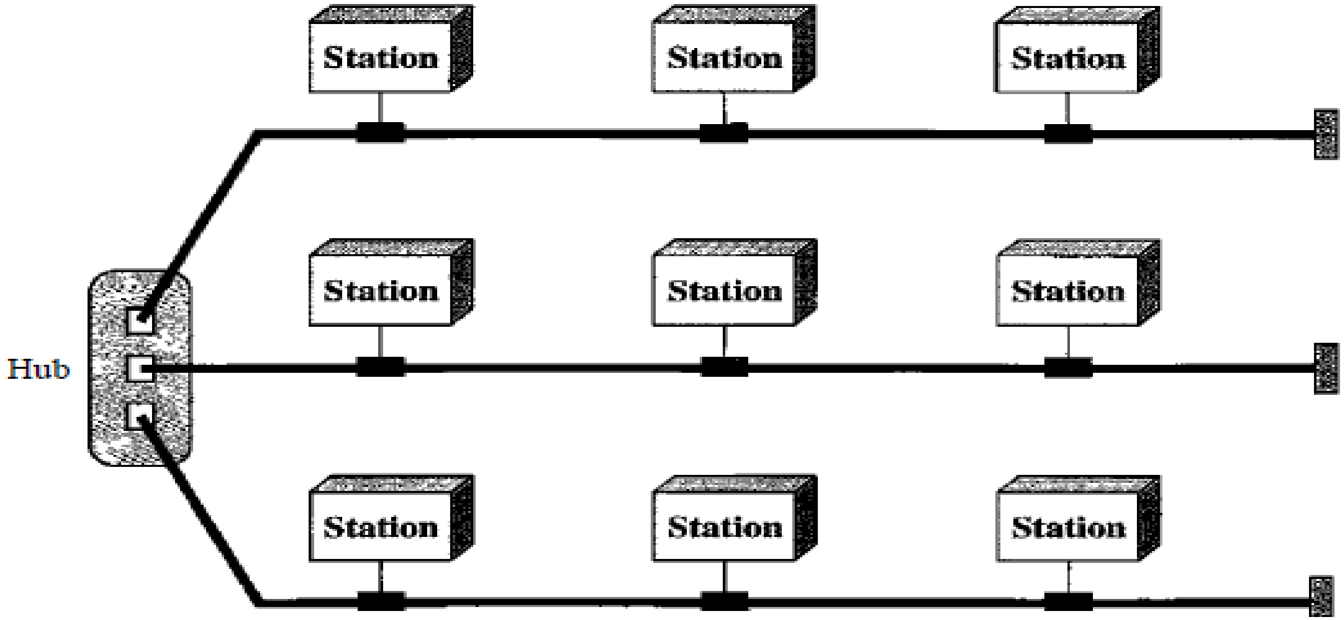
Mesh topology

- In mesh network, there is random connection of nodes using communication links.
- Mesh topology is the general topology for wide area network.
- A mesh network may be fully connected or connected with only partial links.



Hybrid topology

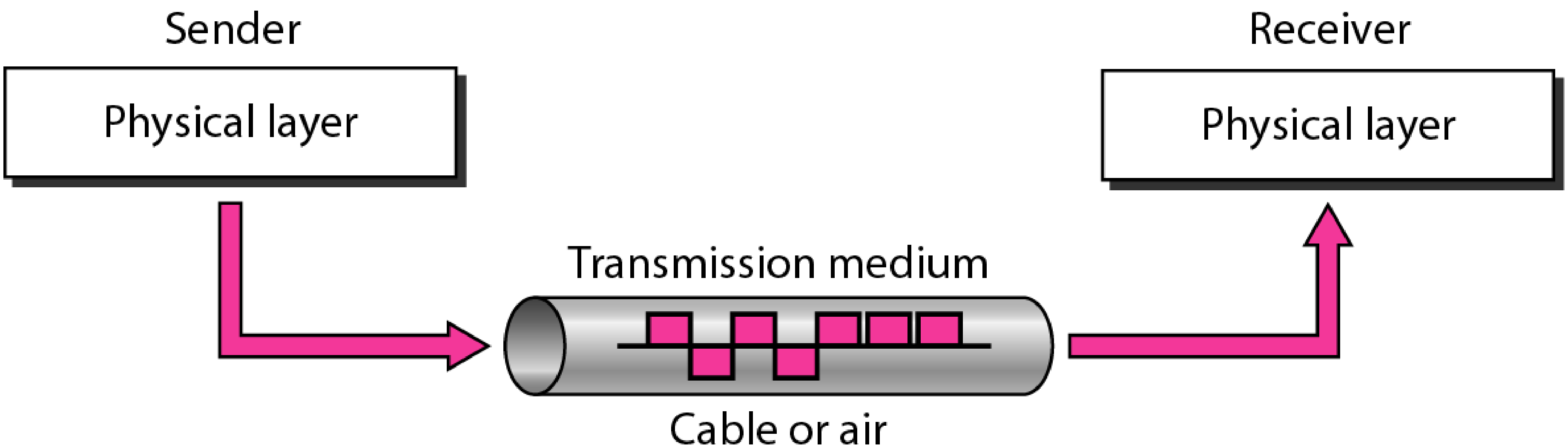
- Mixture of two different topologies.
- This combination of topologies is done according to the requirements of the organization.
- The combined topologies must be different eg. Star-bus or star-ring



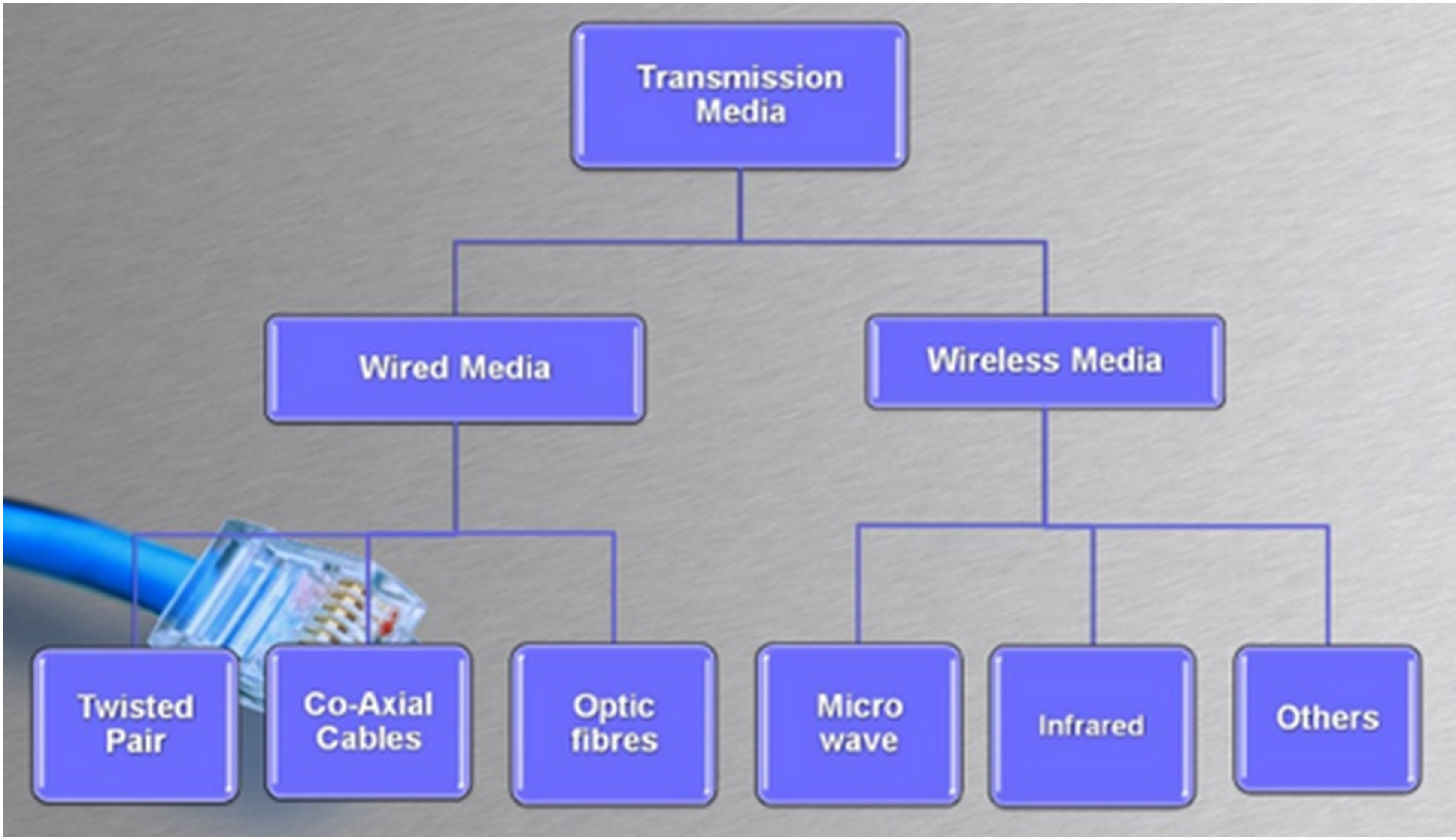
Transmission Media

Transmission Media

- **Transmission media** is a pathway that carries the information from sender to receiver.
- We use different types of cables or waves to transmit data.
- Transmission media is also called **Communication channel.**



Guided and Unguided Transmission Media

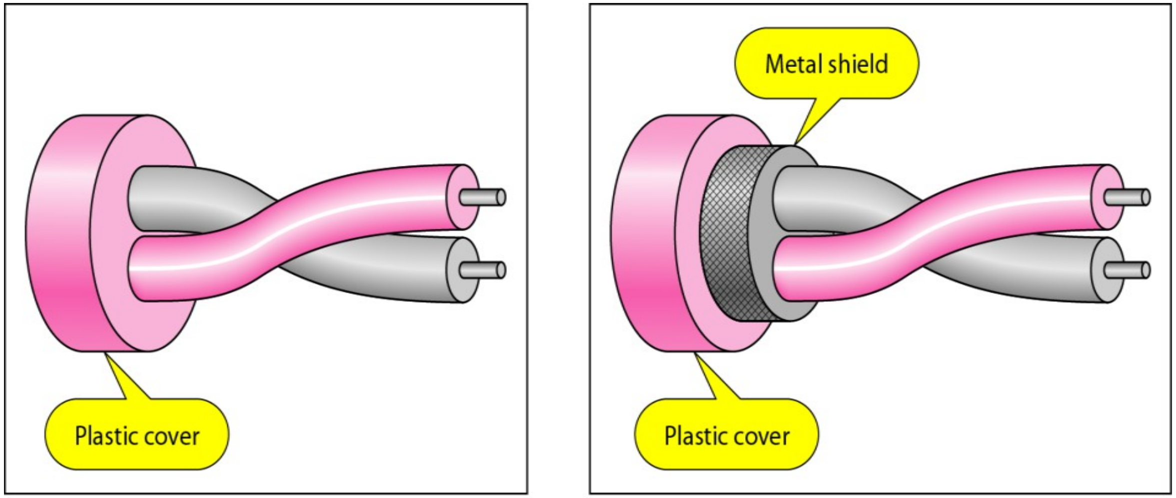


Guided / Wired Transmission Media

- Another name of Wire Transmission Media is called **Bound Transmission Media** or **Guided Transmission Media.**
- Bound transmission media are the cables that are tangible or have physical existence and are limited by the physical geography.
- Popular bound transmission media in use are twisted pair cable, co-axial cable and fiber optical cable.
- Each of them has its own characteristics like transmission speed, effect of noise, physical appearance, cost etc.

Twisted Pair Cable

- **Twisted pair cabling** is a type of wiring in which two conductors of a single circuit are twisted together for the purposes of canceling out electromagnetic interference (EMI) from external sources;
- Twisted pair cables are quite literally a pair of insulated wires that are twisted together to help reduce noise from outside sources.
- Most common transmission wires because of good performance at low cost.



a. UTP b. STP

Cont..

- There are two type of twisted pair cable:
  - Unshielded twisted-pair cables (UTP cables)
  - Shielded twisted-pair cables (STP cables)

Unshielded twisted-pair cables (UTP cables)

- UTP is the most popular type of twisted-pair cable and is fast becoming the most popular LAN cabling.
- Unshielded means no additional shielding like meshes or aluminum foil.
- Due to its low cost, UTP cabling is used extensively for local-area networks (LANs) and telephone connections.
- UTP cabling does not offer a high bandwidth.
- The maximum cable length segment is 100 meters.

Connectors

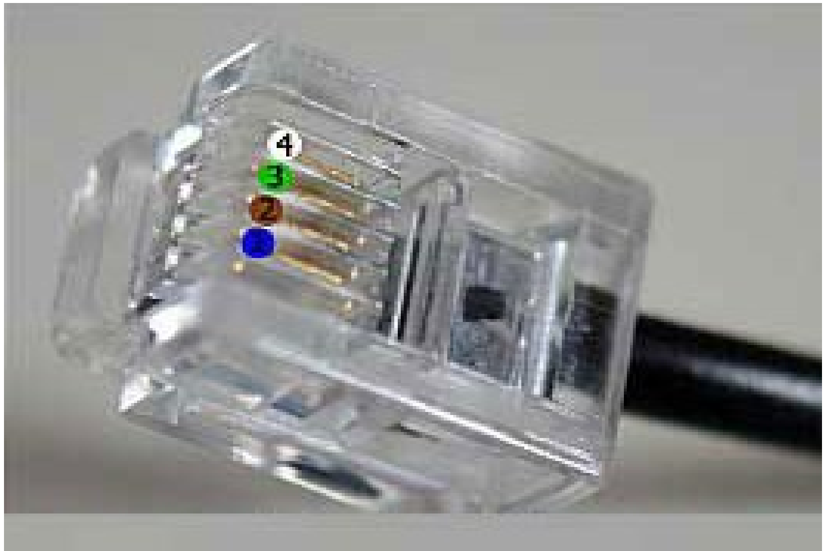
Registered jack (RJ)

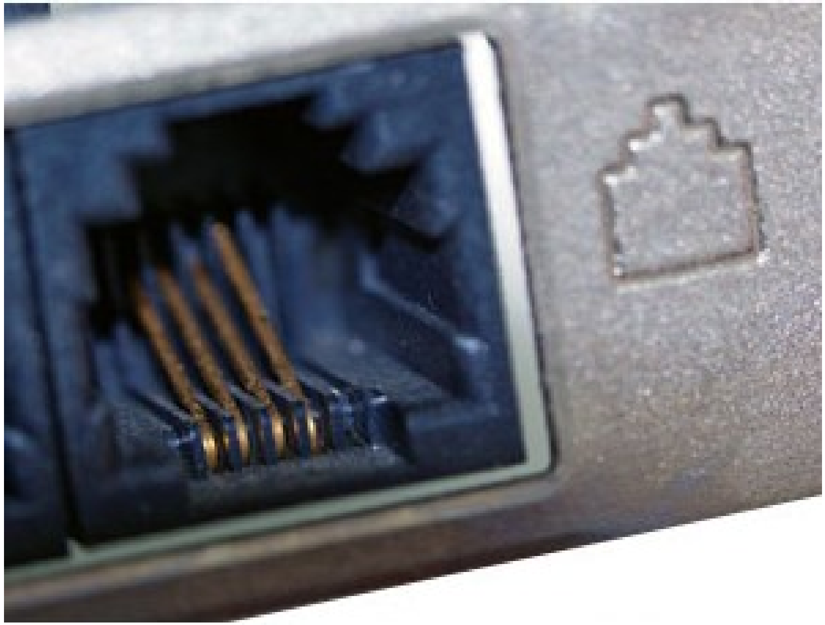
- The most common UTP connector is A registered jack (RJ).
- A registered jack (RJ) is a standardized network interface used for network cabling, wiring and jack construction.
- The primary function of registered jacks is to connect different data equipment and telecommunication devices with services normally provided by telephone exchanges or long-distance carriers.
- Most commonly used RJ

are RJ 11 and RJ 45

Registered Jack - 11

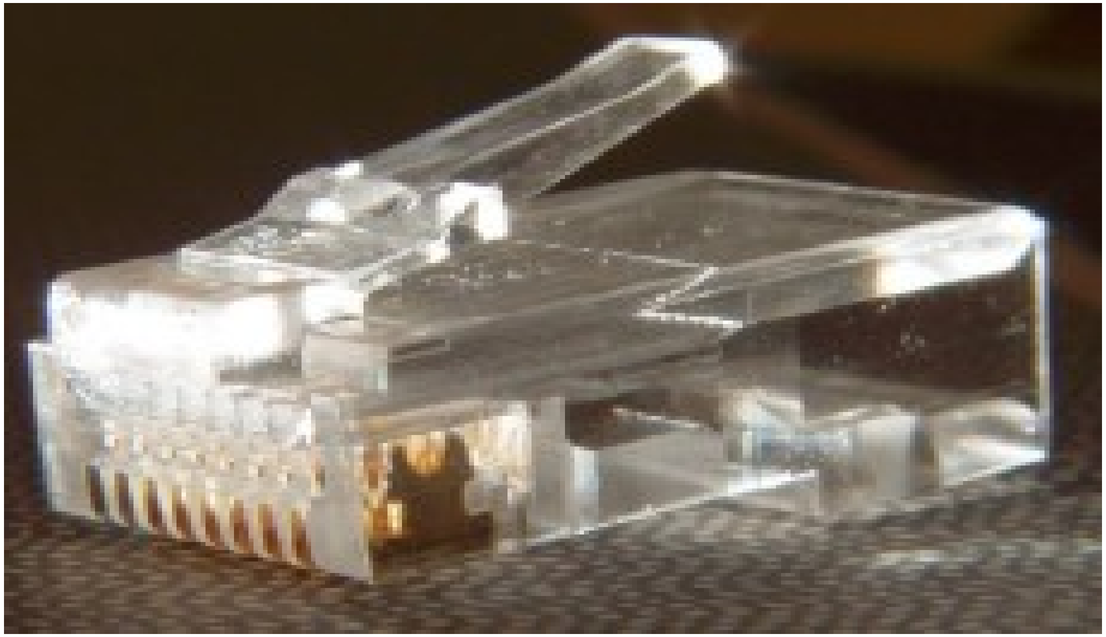
- More commonly known as a **modem port**, **phone connector**, **phone jack** or **phone line**, the **Registered Jack-11** (**RJ-11**) is a for telephone and Modem connectors.
- RJ-11 is a common connector for plugging a telephone into the wall and the handset into the telephone.
- It is found in houses and offices where old telephone-wired systems are connected with the ISP's line.
- It has 2, 4 or 6 pins (contacts)





Registered Jack - 45

- The most common UTP connector is RJ45 (RJ stands for registered jack).
- RJ45 is a standard type of connector for network cables.
- The most common twisted-pair connector is an 8-position, 8-contact (8P8C) modular plug and jack commonly referred to as an RJ45 connector.



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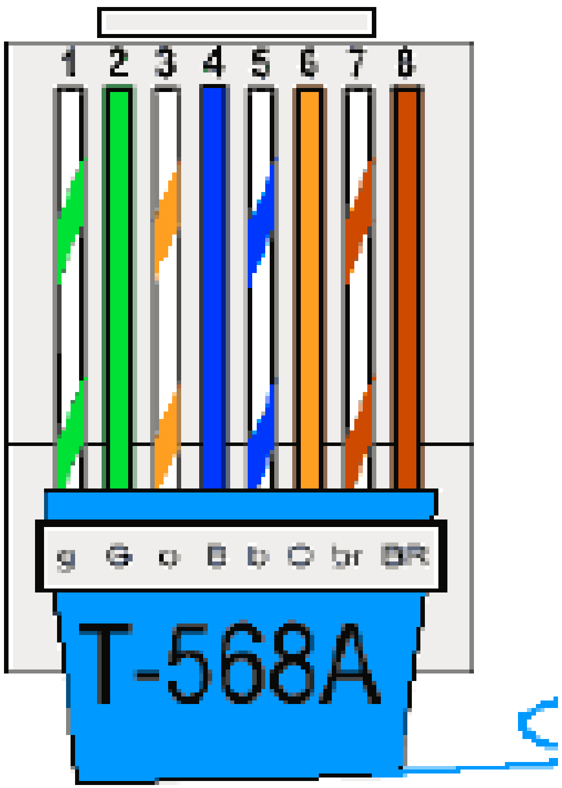
- RJ45 connectors feature eight pins to which the wire strands of a cable interface electrically.
- Each RJ45 connector has eight pins, which means an RJ45 cable contains eight separate wires.
- RJ45 cables can be wired in two different ways. One version is called T-568A and the other is T-568B.

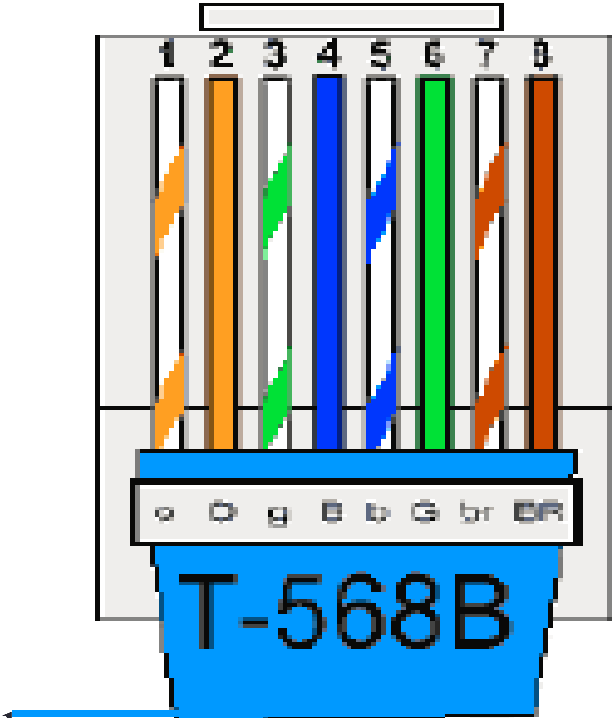
T-568A

1. White/Green (Receive)
2. Green (Receive)
3. White/Orange (Transmit)
4. Blue
5. White/Blue
6. Orange (Transmit)
7. White/Brown
8. Brown



T-568B

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown



- The T-568B wiring scheme is by far … many devices support the T-568A wiring scheme as well.
- Some networking applications require a crossover cable, which has a T-568A connector on one end and a T-568B connector on the other.
- This type of cable is typically used for direct computer-to-computer connections when there is no router, hub, or switch available.

Shielded Twisted Pair (STP)

- Include two individual wires covered with a foil shielding, which prevents electromagnetic interference, thereby transporting data faster.
- STP is similar to unshielded twisted pair (UTP); however, it contains an extra foil wrapping or copper braid jacket to help shield the cable signals from interference.
- STP cables are costlier when compared to UTP, but has the advantage of being capable of supporting higher transmission rates across longer distances.
- The extra covering in shielded twisted pair wiring protects the transmission line from electromagnetic interference leaking into or out of the cable.

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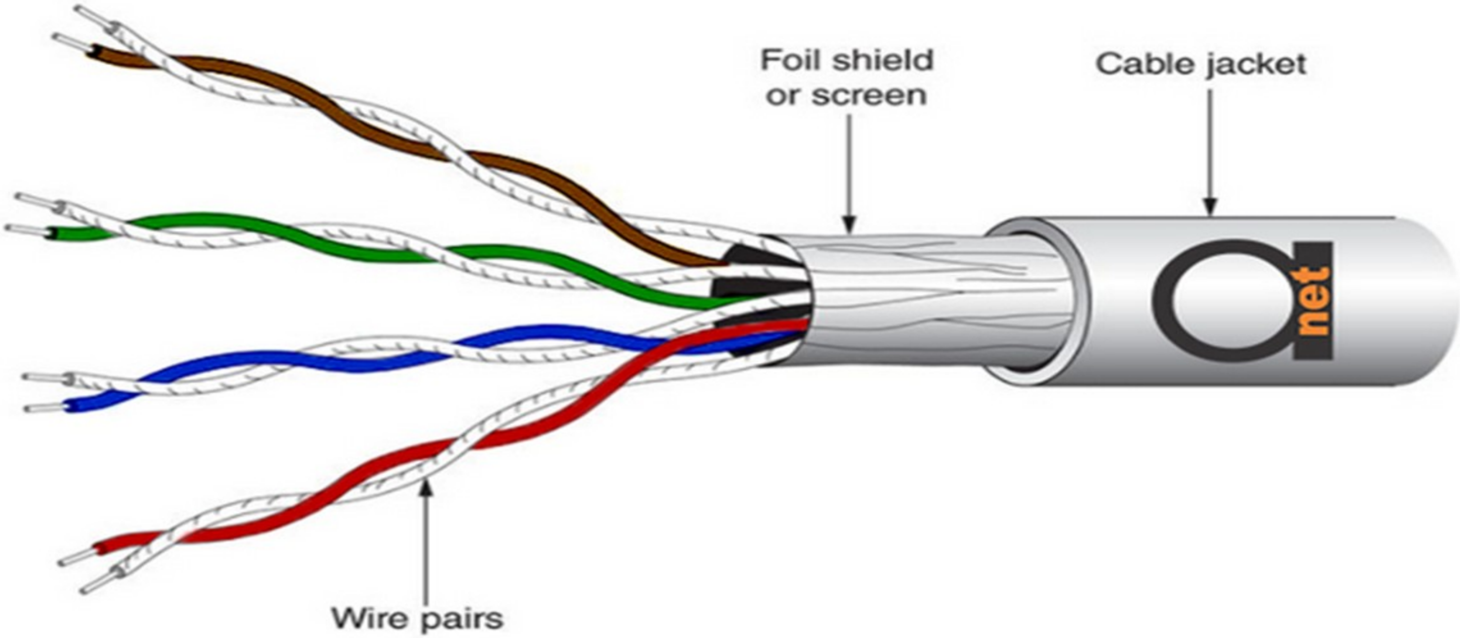
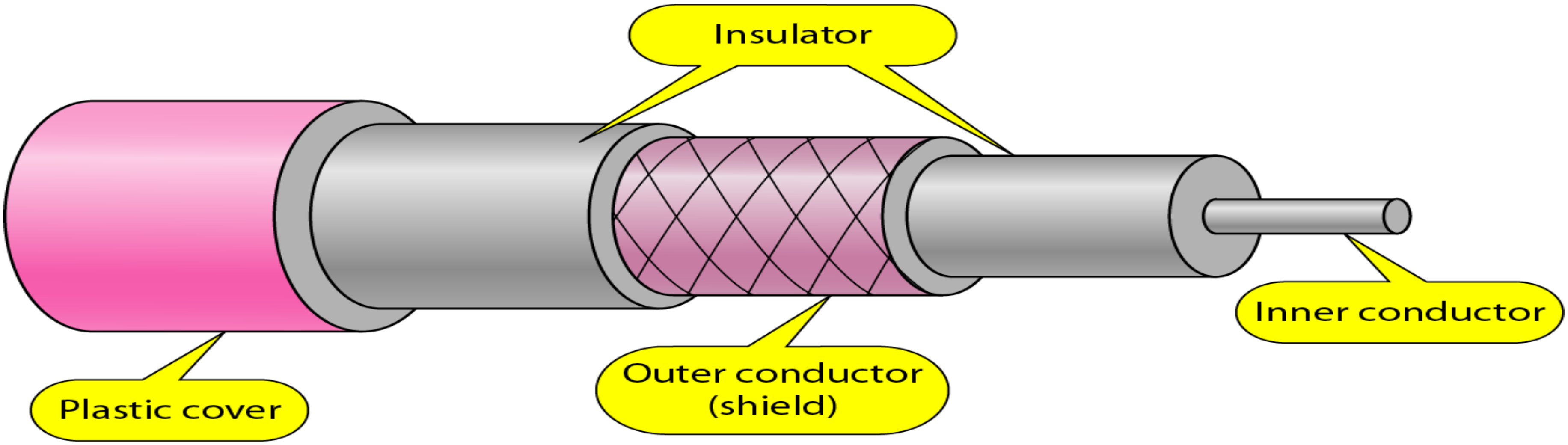
- Mostly used in large industry with high electromagnetic interference.



Coaxial cables

- consists of a core of copper wire surrounded by insulation, a braided metal shielding, and an outer cover.
- has high bandwidths and greater transmission capacity.
- used to connect their TVs to a cable TV service and used in networks and what allow a broadband cable Internet connection using a cable modem.



Fiber optics cable

- contains glass (or in some cases, plastic) fibers rather than copper wire.
- Signals are transmitted across these fibers in the form of light pulses rather than electrical pulses.
- The light signals do not travel at the speed light because of the denser glass layers, instead traveling about 30% slower than the speed of light.
- It is good for very high-speed, high-capacity data transmission because of the purity of the signal and lack of signal attenuation.

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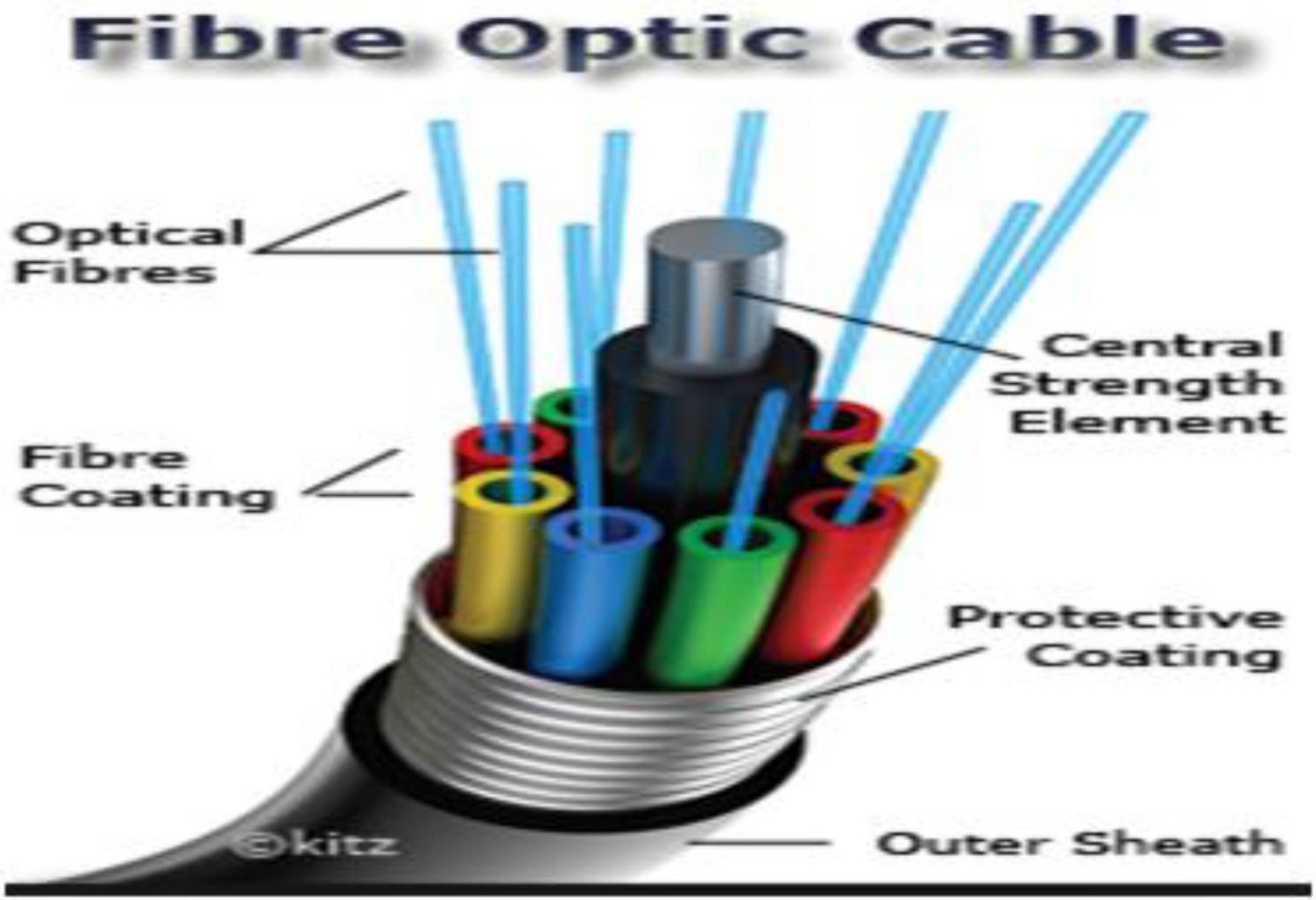

Fibre Optic Cable
Optical
Fibres
Central
Strength
Element
Fibre
Coating
Protective
Coating
©kitz
Outer Sheath

Unguided / Wireless Transmission Media

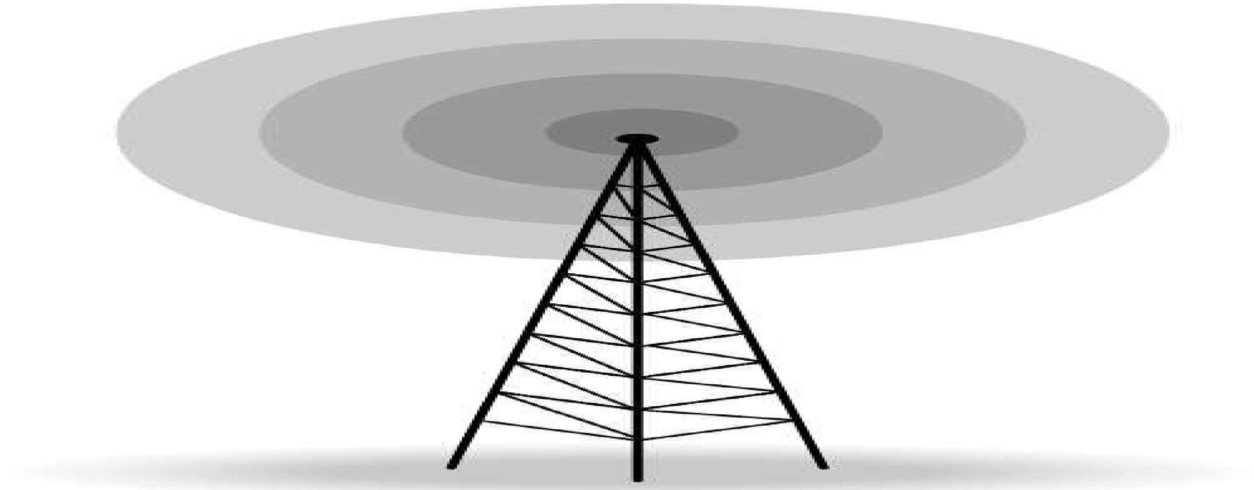
- Unguided media transport electromagnetic waves without using a physical conductor.
- This type of communication is often referred to as wireless communication.
- Signals are normally broadcast through free space and thus are available to anyone who has a device that is capable of receiving them.

Wireless LAN

- Wireless communication is one of the fastest-growing technologies.
- The demand for connecting devices without the use of cables is increasing everywhere.
- It is a wireless computer network that links two or more devises using a wireless communication to form a local area network (LAN).
- Wireless LANs can be found in college campuses, in office buildings.

Radio Communication

- Uses electromagnetic waves with the longest wavelengths.
- Used for communication; television, and radios etc.
- All receive radio waves and convert them to mechanical vibrations in the speaker to create sound waves that can be heard.
- Used as a multicast data transmission. there is one sender but many receivers. FM radio, television,



Microwave Communication

- Used for long-distance telephone service. It uses radio frequency for signal transmissions.
- The microwave band is well suited for wireless transmission of signals having large bandwidth.
- Microwaves are unidirectional that means the communication is point to point.
- When an antenna transmits microwave waves, they can be narrowly focused. This means that the sending and receiving antennas need to be aligned.
- Eg. Used for Television distribution, Long-distance telephone transmission and Private business networks

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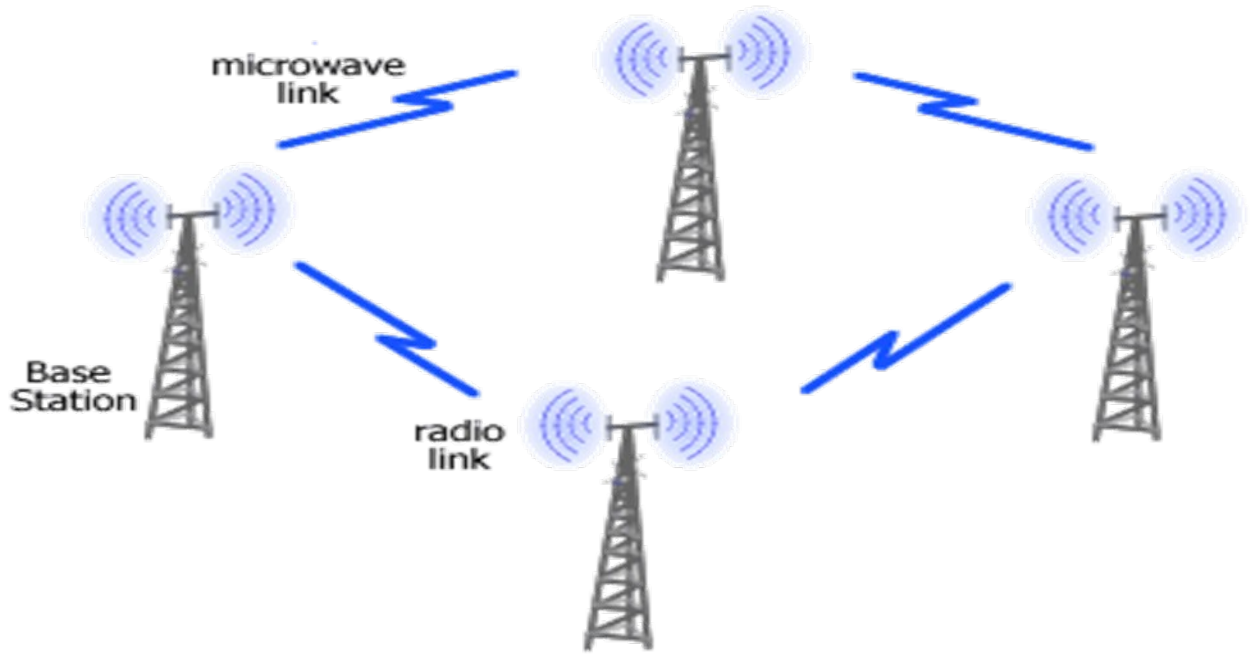


Infrared Communication

- The frequency of light that is not visible to human eyes.
- Infrared can be used for short-range communication.
- Infrared communication requires a transceiver (a combination of transmitter and receiver) in both devices that communicate.
- It has high frequencies, shorter wavelengths and cannot penetrate walls.
- Eg. Remote controls for TVs

Wi-Fi

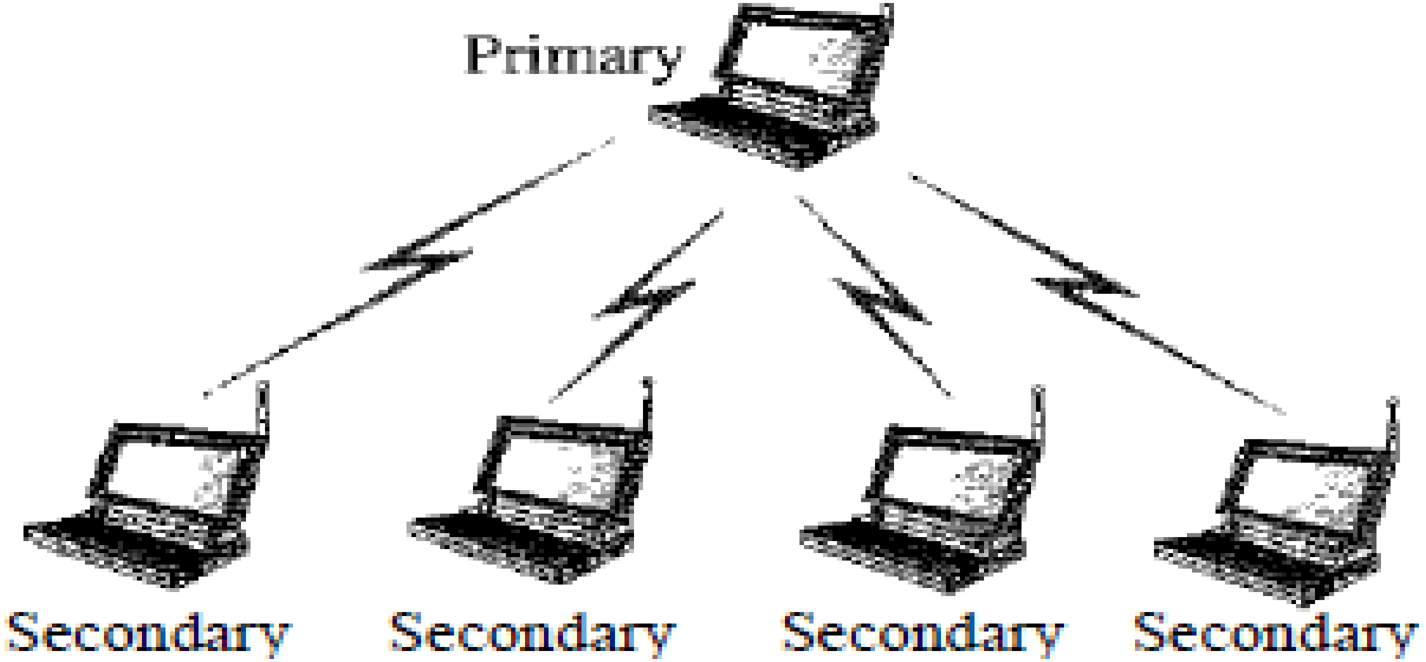
- Wi-Fi is a wireless technology that is used to connect computers, tablets, smartphones and other devices to the internet.
- It uses radio waves to transmit data from your wireless router to your Wi-Fi enabled devices such as Smartphones, tablets, computers.

Bluetooth Technology

- Bluetooth is a wireless LAN technology designed to connect devices of different functions such as telephones, notebooks, computers (desktop and laptop), cameras, printers, coffee makers, and so on.
- A Bluetooth LAN, by nature, cannot be large.
- Bluetooth technology has several applications.
  - Peripheral devices such as a wireless mouse or keyboard can communicate with the computer through this technology.
  - Monitoring devices can communicate with sensor devices in a small health care center.

Cont..

- Home security devices can use this technology to connect different sensors to the main security controller.
- Conference attendees can synchronize their laptop computers at a conference.



Satellite communication

- Satellite communication is a special use of microwave transmission system.
- The ground station consists of a satellite dish that functions as an antenna and communication equipment to transmit (called Uplink) and receive (called Downlink) data from satellites passing overhead.
- Satellites are especially used for remote locations, which are difficult to reach with wired infrastructure.
- Hence it always stays over the same point with respect to the earth.
- This allows the ground station to aim its antenna at a fixed point in the sky.

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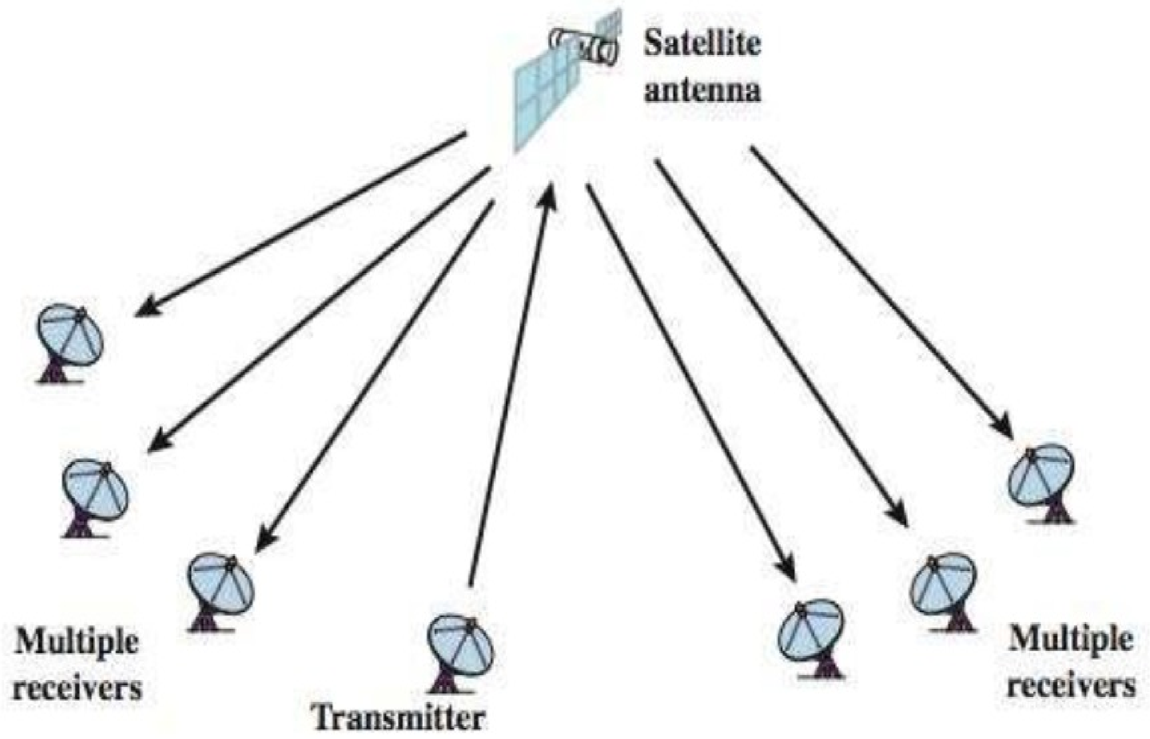

(b) Broadcast link

Protocols and standards

- For communication to occur, the entities must agree on a **protocol**.
- A protocol is a set of rules that govern data communications.
- A protocol defines what is communicated, how it is communicated, and when it is communicated.
- The key elements of a protocol are **syntax**, **semantics**, and **timing**.
- ***Syntax*** refers to the structure or format of the data, meaning the order in which they are presented.
- For example, a simple protocol might expect the first 8 bits of data to be the address of the sender, the second 8 bits to be the address of the receiver, and the rest of the stream to be the message itself.

Standards

- Standards provide guidelines to manufacturers, vendors, government agencies, and other service providers to ensure the kind of interconnectivity necessary in today's marketplace and in international communications.
- Standards are essential in creating and maintaining an open and competitive market for equipment manufacturers and in guaranteeing national and international interoperability of data and telecommunications technology and processes.

THANK YOU!!!